

Heterogeneity and Aggregate Fluctuations

Minsu Chang

Seoul National University

Xiaohong Chen

Yale University and Cowles Foundation

Frank Schorfheide

University of Pennsylvania, Centre for Economic Policy Research, Penn Institute for Economic Research, and National Bureau of Economic Research

We develop a state-space model with a transition equation that takes the form of a functional vector autoregression (VAR) and stacks macroeconomic aggregates and a cross-sectional density. The measurement equation captures the error in estimating log densities from repeated cross-sectional samples. The log densities and their transition kernels are approximated by sieves, which leads to a finite-dimensional VAR for macroeconomic aggregates and sieve coefficients. With this model, we study the dynamics of technology shocks, GDP (gross domestic product), employment, and the earnings distribution. We find that spillovers between aggregate and distributional dynamics are generally small, that a positive technology shock tends to decrease inequality, and that a shock that raises earnings inequality leads to a small and insignificant GDP response.

We thank Bence Bardoczy, Yongsung Chang, Stephanie Ettmeier, Mark Huggett, Dirk Krüger, Víctor Ríos-Rull, Tom Sargent, Harald Uhlig, two anonymous referees, and participants at various seminars and conferences for helpful suggestions. Schorfheide gratefully acknowledges financial support from the National Science Foundation under grant SES 1851634. This paper was edited by Harald Uhlig.

Electronically published November 4, 2024

Journal of Political Economy, volume 132, number 12, December 2024.

© 2024 The University of Chicago. All rights reserved. Published by The University of Chicago Press.

<https://doi.org/10.1086/731411>

I. Introduction

Models with household or firm heterogeneity are widely used to study distributional effects of macroeconomic policies. In these models, heterogeneity evolves dynamically and potentially interacts closely with macroeconomic aggregates such as GDP (gross domestic product), unemployment, and investment. However, it is an open question to what extent model predictions are consistent with empirical evidence. This paper develops a state-space model that can provide semistructural evidence about the interaction of aggregate and distributional dynamics at business-cycle frequencies. The state-transition equation takes the form of a linear functional vector autoregression (VAR), stacking macroeconomic aggregates and the log density of a cross-sectional distribution. We treat this log density as unobserved and specify a measurement equation for unit-level cross-sectional observations. Our model can be used to trace out the effects of aggregate shocks on cross-sectional distributions; assess whether fluctuations in cross-sectional distributions are important for aggregate fluctuations, in the sense of Granger causality; and identify shocks to cross-sectional distributions and examine their effects on aggregate variables.

We represent the log densities and the transition kernels in the functional VAR by finite-dimensional sieves with fixed basis functions and time-varying coefficients that capture the dynamics. This turns the state-transition equation into a linear VAR for the aggregate variables and the time-varying sieve coefficients. To avoid nonlinear filtering, we effectively linearize the measurement equation for the cross-sectional observation, which leads to a convenient two-step estimation procedure. First, one computes maximum likelihood estimates (MLEs) of the sieve coefficients from the cross-sectional data, separately for each time period. Second, one estimates a linear state-space model in which the MLEs summarize the cross-sectional data and serve as noisy measures of the sieve coefficients. We implement the second step with Bayesian techniques. A novel feature of our two-step approach is its ability to capture estimation uncertainty about the unobserved cross-sectional densities. For data-driven selection of the sieve dimension, the number of VAR lags, and hyperparameters that control the variance of the prior distribution, we derive an approximation of the Bayesian marginal data density (MDD).

In addition to being able to capture density-estimation uncertainty, there are multiple benefits associated with our functional approach. First, we model the entire cross-sectional distribution, and not just particular quantiles or inequality measures. Once that has been done, any distributional statistic can be computed by postprocessing the initial density estimates. This is particularly useful when users start out from a broad research question—for example, How does a structural shock affect earnings inequality?—instead of a narrow question such as how the 10th and 90th percentiles

respond to an aggregate shock. Second, our representation of the cross-sectional distributions through unnormalized log densities can be coherently embedded in a linear autoregressive framework. Unlike quantiles, cumulative distribution functions, or probability density functions, the unnormalized log densities do not have to obey monotonicity or nonnegativity that cannot be enforced easily in a linear law of motion.

Third, the framework facilitates the identification of structural shocks in multiple dimensions. In our empirical application, we study the propagation of novel shocks that are identified by their effect on the cross-sectional distribution. Although not explored in this paper, the inclusion of a cross-sectional distribution in a vector autoregressive law of motion may also assist the shock identification by ensuring that the VAR innovations span the structural shocks—for example, if a policy maker systematically responds to fluctuations in the cross-sectional distribution—or by generating new sign restrictions that reduce the size of identified sets, for example, along the lines of Amir-Ahmadi and Drautzburg (2021).

A prospective user may be concerned that the implementation costs of the proposed functional approach are substantially higher than including a few cross-sectional statistics, such as percentiles, in a VAR. One of the contributions of our paper is to show that this is not the case. If the user chooses to ignore the uncertainty associated with the cross-sectional density estimates, which is the *modus operandi* in VARs that are augmented by cross-sectional statistics, then the method amounts to stacking macro variables and sieve-coefficient MLEs and estimating a VAR with the augmented vector. Our uncertainty-quantification step can be easily implemented by estimating a linear Gaussian state-space model instead of a VAR. In addition to software that replicates the simulations and empirical analysis in this paper, we provide a simplified set of codes that implements the first step of the analysis (cross-sectional density estimation) and can be combined with existing VAR and state-space-model-estimation software.

We present two asymptotic results, assuming that the number of sieve dimensions K , the cross-sectional sample size N , and the time series length T tend to infinity. Under the assumption that $K^3/N = o(1)$, our first theorem shows that the difference between the exact likelihood and the likelihood under the linearized measurement equation vanishes. This result is confirmed in a computational exercise in which we use an approximately conditionally optimal particle filter (see Herbst and Schorfheide 2015) to numerically evaluate the exact likelihood function of the K -dimensional nonlinear state-space model. Because the error from linearizing the measurement equation is small and the run time is reduced by orders of magnitude, we recommend the linearization for empirical work. Our second theorem shows that the discrepancy between the exact MDD and our MDD approximation (used for data-driven model selection) also vanishes, provided that $KT/N = o(1)$ holds in addition. As a corollary, by maximizing

the approximate MDD we also “almost” achieve the maximum value for the exact MDD. Again, we think that the computational gain (in terms of time and precision) from working with an approximation outweighs potential distortions from the approximation error in empirical work.

In a simulation experiment, we generate data from a version of the Krusell and Smith (1998) model, henceforth the “KS model,” which we solve with the method proposed by Winberry (2018). The estimates of our functional model can reproduce the evolution of the distribution of asset holdings and the response of the cross-sectional distribution to a technology shock. The MDD shrinks to zero the coefficients that control the spillover from the lagged cross-sectional distribution to current aggregate capital. Thus, according to the estimated functional model, the cross-sectional distribution does not Granger-cause the aggregate capital stock. This finding is a manifestation of the well-known aggregation property of the KS model, which holds exactly in our version of the model. Because of the absence of Granger causality, the estimated response of capital to a technology shock is identical to that from a VAR in the two aggregate variables only, using a single lag as in the functional model.

We conduct a second simulation with a modified KS model in which we break the perfect-aggregation property and the cross-sectional distribution Granger-causes aggregate capital. In this case, the MDD does not shrink the spillover coefficients to zero, and, with the lag length in both models fixed at 1, there are discrepancies between the functional and aggregate VAR impulse-response functions (IRFs) of capital. Because the functional model can be rewritten as an infinite-order vector autoregressive system in terms of the two aggregate variables, increasing the number of lags in the aggregate VAR makes this discrepancy vanish.

In an empirical application, we fit our model to aggregate total factor productivity (TFP) growth, GDP growth, employment, and cross-sectional data on labor earnings divided by GDP per capita. We obtain the following results. First, on the basis of the degree of shrinkage selected with the MDD criterion, the Granger-causal relationship from the cross-sectional income distribution to the aggregate variables is weak. Estimated impulse responses of the macro variables based on the functional model and an aggregate VAR with the same lag length are very similar, mirroring the simulation results from the baseline KS economy. Second, the earnings density responds to a TFP shock in such a way that inequality falls. The implied 10th percentile and the fraction of individuals earning less than per capita GDP rise, whereas normalized earnings at the 50th and 90th percentiles and the Gini coefficient fall. Overall, the effects are quantitatively small and largely driven by individuals transitioning between work and unemployment.

Third, the proposed functional model delivers sharper estimates of percentile and inequality measure IRFs than a VAR that combines these

statistics directly with the aggregate variables.¹ We obtain the same result using data simulated from the KS economy. Our functional approach also guarantees that percentiles do not cross in response to shocks and in a forward simulation of the model. Fourth, using maximum-share-of-variance identification schemes, we find that IRFs from shocks that maximize the variance share of GDP or employment look very similar to TFP IRFs. Distributional shocks that explain the maximal variance share in the overall cross-sectional distribution or the Gini coefficient do not have a significant effect on GDP.²

The methods developed in this paper speak to some but not all aspects of the broader question of whether heterogeneity matters for business-cycle fluctuations. The functional model generates direct estimates for responses of cross-sectional distributions to aggregate shocks and for the effects of shocks to the cross-sectional distribution on aggregate outcomes. Quantitative HA (heterogeneous-agent) models can be evaluated on their ability to replicate these empirical IRFs. Assessing the role that heterogeneity plays for the propagation of aggregate shocks on aggregate outcomes is more delicate. Heterogeneity may matter because it affects the mapping from deep technology and preference parameters into the VAR representation. Our functional model cannot unveil this mapping; it simply generates estimates of the VAR parameters that track the observed business-cycle fluctuations.³ The proposed method can, however, be used to assess whether cross-sectional distributions Granger-cause aggregates and to examine discrepancies between IRFs from the estimated functional model and those from a VAR that uses only aggregate data. A well-fitting quantitative HA model will have to reproduce the empirical Granger causality and IRF mismatch patterns.

The strong overlap between estimated IRFs from the functional model and an aggregate VAR is consistent with the HA and RA New Keynesian model-estimation results reported in Bayer, Born, and Luetticke (2024). It is also consistent with the notion that many HA models deliver approximate aggregation results; see Chang, Kim, and Schorfheide (2013), Werning (2015), and Berger, Bocola, and Dovis (2023). The latter paper finds that the role of state-dependent preference shocks, which summarize all the information from the cross section relevant for aggregate dynamics, is quantitatively small. Bilbiie, Primiceri, and Tambalotti (2023) estimate

¹ An example of this approach is Coibion et al. (2017).

² Identification based on the maximum share of variance of an observable dates back to Faust (1998) and Uhlig (2003) and has recently been applied by Angeletos, Collard, and Dellas (2020) to study what those authors refer to as the “anatomy” of business cycles.

³ There is no analogue in our framework to the popular experiment of comparing HA and RA (representative-agent) IRFs by shutting down cross-sectional heterogeneity as in, for instance, Krueger, Mitman, and Perri (2016), Ahn et al. (2018), Kaplan and Violante (2018), Ottonello and Winberry (2020), Villalvazo (2021), Cho (2023) and Bayer, Born, and Luetticke (2024).

the amplification of business-cycle fluctuations due to precautionary savings in a medium-scale New Keynesian model with savers and hand-to-mouth households. The estimated amplification is observationally similar to more volatile shocks in an RA environment. This suggests that their framework will also be consistent with the strong overlap between IRFs from the functional model and the aggregate VAR that we document.

HA models are frequently used to study the effect of aggregate shocks on cross-sectional distributions. For instance, the paper by Ahn et al. (2018) studies the effect of factor-specific productivity shocks on inequality dynamics, whereas Kaplan and Violante (2018) examine the distributional effects of monetary policy shocks. Bayer, Born, and Luetticke (2024) use their estimated HA model to construct a historical decomposition of an inequality measure with respect to a collection of aggregate shocks. Bhandari et al. (2021) report responses of the dispersion of assets to a TFP shock in a HANK (HA New Keynesian) model under optimal monetary-fiscal policy. Mongey and Williams (2017) analyze the effect of macro shocks on the dispersion of sales growth. Examples of research examining the effect of distributional shocks on aggregate variables are papers by Huggett (1997) and Auclert and Rognlie (2020). The former shows that a redistribution of asset holdings among households while keeping the overall capital stock fixed at its steady-state level triggers a response of the aggregate variables. Auclert and Rognlie (2020) show that a rise in inequality may lower aggregate output if monetary policy does not react to it.

The structure of the transition equation in our functional model resembles that of HA models solved with linearization techniques. This solution method was initially proposed by Reiter (2009); has been further developed in several papers, including Childers (2018), Kaplan and Violante (2018), and Winberry (2018); and is often used for the likelihood-based estimation of HA models, as in Mongey and Williams (2017), Acharya et al. (2019), Cho (2023), Liu and Plagborg-Møller (2023), and Bayer, Born, and Luetticke (2024). In terms of likelihood construction, the Liu and Plagborg-Møller (2023) paper is most closely related to ours, because it also uses the density of cross-sectional observations. However, in their case the density is part of the HA model solution, whereas we estimate it flexibly from the data.

There is an extensive literature on the statistical analysis of functional data. General treatments are provided in the books by Bosq (2000), Ramsey and Silverman (2005), and Horváth and Kokoszka (2012). Unlike the majority of papers, we do not assume that the functions are observed without error. Applications of functional data analysis in macroeconometrics are growing steadily. Many of them are related to the yield curve, for example, Diebold and Li (2006) and Inoue and Rossi (2021). Meeks and Monti (2023) use functional principal component regression to estimate a New Keynesian Phillips curve in which the distribution of inflation expectations

appears on the right-hand side. Hu and Park (2017) develop an estimation theory for a functional autoregressive model with unit roots and fit it to yield-curve data, and Chang, Kim, and Park (2016) use a functional time-series process to capture the evolution of earnings densities with a focus on unit-root components. Both papers use functional principal components analysis.

The remainder of this paper is organized as follows. In section II, we present our functional state-space model for a group of macroeconomic time series and a sequence of cross-sectional distributions. We develop an approximate filter for the state-space model that facilitates the likelihood-based estimation. We use Bayesian inference and derive an approximation of the MDD that is used for dimensionality and hyperparameter selection. Theoretical results on the accuracy of the approximations are presented in section III. Implementation details such as the choice of basis functions, the choice of prior distributions, and the posterior sampler are discussed in section IV. Section V contains the results from the simulation experiment in which we estimate the functional state-space model based on data generated from the KS economy. The empirical application is presented in section VI, and section VII concludes. Supplemental derivations and additional computational details and empirical results are relegated to the appendix, available online.⁴

II. A Functional State-Space Model

We now develop the functional model, starting from an infinite-dimensional specification in section II.A. Rather than working with an infinite-dimensional model, we consider a collection of finite-dimensional models, presented in section II.B, in which log densities and kernels associated with integral operators are represented through finite-dimensional sieves. Section II.C develops an approximate linear filter for the finite-dimensional functional state-space models, and in section II.D we propose a large-sample approximation of the MDD that can be used to select the degree of the sieve approximation, the number of lags, and prior hyperparameters in a data-driven manner.

A. An Infinite-Dimensional Model

To fix ideas, we start with an infinite-dimensional model for an $n_y \times 1$ vector of macroeconomic aggregates Y_t and a cross-sectional density $p_t(x)$. In our empirical application, Y_t consists of (log) TFP growth, per capita GDP

⁴ Julia scripts to replicate the simulation experiment and the empirical analysis and to combine the functional analysis with existing VAR and state-space-model-estimation software are available from the authors' websites.

growth, and the log employment rate. The cross-sectional variable x represents earnings as a fraction of per capita GDP. Throughout this paper, we work with log densities defined as $\ell_t(x) = \ln p_t(x)$. The advantage of log densities is that they are not subject to nonnegativity or monotonicity restrictions that would be difficult to enforce in a linear vector autoregressive law of motion. We decompose Y_t and ℓ_t into a deterministic component $(Y_*, \ell_*(x))$ and fluctuations around it. Let

$$Y_t = Y_* + \tilde{Y}_t \quad \text{and} \quad \ell_t = \ell_* + \tilde{\ell}_t. \quad (1)$$

For notational convenience, we assume that the deterministic component is time invariant and could be interpreted as a steady state. This assumption could be easily relaxed by letting (Y_*, ℓ_*) depend on t . The deviations from the deterministic component $(Y_t, \ell_t(x))$ evolve jointly according to the following linear functional VAR:

$$\begin{aligned} \tilde{Y}_t &= B_{yy} \tilde{Y}_{t-1} + \int B_{yt}(\tilde{x}) \tilde{\ell}_{t-1}(\tilde{x}) d\tilde{x} + u_{y,t}, \\ \tilde{\ell}_t(x) &= B_{ly}(x) \tilde{Y}_{t-1} + \int B_{ll}(x, \tilde{x}) \tilde{\ell}_{t-1}(\tilde{x}) d\tilde{x} + u_{l,t}(x). \end{aligned} \quad (2)$$

Here $u_{y,t}$ is mean-zero random vector with covariance Ω_{yy} and $u_{l,t}(x)$ is a random element in a Hilbert space with covariance function $\Omega_{ll}(x, \tilde{x})$. We denote the covariance function for $u_{y,t}$ and $u_{l,t}(x)$ by $\Omega_{yl}(x)$. For now, equation (2) should be interpreted as a reduced-form functional VAR in which $u_{y,t}$ and $u_{l,t}(x)$ are one-step-ahead forecast errors. The time t density for the cross-sectional observations x_{it} is given by

$$p_t(x) = \frac{\exp\{\ell_*(x) + \tilde{\ell}_t(x)\}}{\int \exp\{\ell_*(\tilde{x}) + \tilde{\ell}_t(\tilde{x})\} d\tilde{x}}. \quad (3)$$

The numerator ensures that the density $p_t(x)$ integrates to 1, which is not automatically guaranteed by the linear law of motion in equation (2).

B. A Collection of Finite-Dimensional Models

Instead of working with the infinite-dimensional model comprising equations (1)–(3), we estimate a K -dimensional specification, where K is determined in a data-driven manner.⁵ Let

⁵ We discuss in the appendix how a finite-dimensional approximation can be derived from the infinite-dimensional specification. In this derivation, given the dimension K , the α_i 's minimize the Kullback-Leibler divergence from the infinite-dimensional density.

$$\ell_t^{(K)}(x) = \sum_{k=1}^K \alpha_{k,t} \zeta_k(x) = [\zeta_1(x), \dots, \zeta_K(x)] \cdot \begin{bmatrix} \alpha_{1,t} \\ \vdots \\ \alpha_{K,t} \end{bmatrix} = \zeta'(x) \alpha_t, \quad (4)$$

$\ell_*^{(K)}(x) = \zeta'(x) \alpha_*$, and $\tilde{\ell}_t^{(K)}(x) = \zeta'(x) \tilde{\alpha}_t$. To simplify the notation, we do not use (K) superscripts for the vectors $\zeta(x)$, α_t , and α_* . Here, $\zeta_1(x)$, $\zeta_2(x)$, ... is a sequence of basis functions. For theoretical considerations it is convenient to demean the vector of basis functions and assume that $\int \zeta(x) dx = 0$. For applications, this normalization is not important.

We represent the kernels $B_{ll}(x, \tilde{x})$ and $B_{yl}(\tilde{x})$, the function $B_{ly}(x)$, and the functional innovation $u_{lt}(x)$ that appear in the state-transition equation (2) as follows:

$$\begin{aligned} B_{ll}^{(K)}(x, \tilde{x}) &= \zeta'(x) B_{ll} \xi(\tilde{x}), & B_{yl}^{(K)}(\tilde{x}) &= B_{yl} \xi(\tilde{x}), \\ B_{ly}^{(K)}(x) &= \zeta'(x) B_{ly}, & u_{lt}^{(K)}(x) &= \zeta'(x) u_{\alpha,t}, \end{aligned} \quad (5)$$

where $\xi(x)$ is a second $K \times 1$ vector of basis functions and $u_{\alpha,t}$ is a $K \times 1$ vector of innovations. The matrix B_{ll} is of dimension $K \times K$, B_{yl} is of dimension $n_y \times K$, and B_{ly} is of dimension $K \times n_y$. Plugging equation (5) into equation (2) and assuming that the innovations $u'_t = [u'_{y,t}, u'_{\alpha,t}]$ are Gaussian yields the following vector autoregressive system for the macroeconomic aggregates and the sieve coefficients (omitting K superscripts):

$$\begin{bmatrix} Y_t - Y_* \\ \alpha_t - \alpha_* \end{bmatrix} = \begin{bmatrix} \Phi_{yy} & \Phi_{y\alpha} \\ \Phi_{\alpha y} & \Phi_{\alpha\alpha} \end{bmatrix} \begin{bmatrix} Y_{t-1} - Y_* \\ \alpha_{t-1} - \alpha_* \end{bmatrix} + \begin{bmatrix} u_{y,t} \\ u_{\alpha,t} \end{bmatrix}, \quad u_t \sim \mathcal{N}(0, \Sigma), \quad (6)$$

where $\Phi_{yy} = B_{yy}$, $\Phi_{y\alpha} = B_{yl} C_\alpha$, $\Phi_{\alpha y} = B_{ly}$, $\Phi_{\alpha\alpha} = B_{ll} C_\alpha$, and $C_\alpha = \int \xi(\tilde{x}) \zeta'(\tilde{x}) d\tilde{x}$.

We assume that in every period $t = 1, \dots, T$ an econometrician observes the macroeconomic aggregates Y_t as well as a sample of N draws x_{it} , $i = 1, \dots, N$, from the K th-order cross-sectional density $p_t^{(K)}(x)$:⁶

$$x_{it} | \alpha_t \sim \text{i.i.d. } p_t^{(K)}(x) = \frac{\exp\{\ell_t^{(K)}(x)\}}{\int \exp\{\ell_t^{(K)}(\tilde{x})\} d\tilde{x}}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (7)$$

Write the joint density of the x_{it} 's as the product of the marginals, and let $\mathcal{L}^{(K)}(\alpha_t | X_t) = (1/N) \sum_{i=1}^N \ell_t^{(K)}(x_{it})$. Then, the measurement equation for the time t cross-sectional observations $X_t = [x_{1t}, \dots, x_{Nt}]'$ can be expressed as

⁶ In practice, N may vary with t . It is straightforward to adjust the formulas accordingly.

$$p^{(K)}(X_t|\alpha_t) = \exp\{N\mathcal{L}^{(K)}(\alpha_t|X_t)\}, \quad \mathcal{L}^{(K)}(\alpha_t|X_t) = \bar{\zeta}'(X_t)\alpha_t - \varphi(\alpha_t),$$

$$\bar{\zeta}'(X_t) = \frac{1}{N} \sum_{i=1}^N \zeta'(x_{it}), \quad \varphi(\alpha_t) = \ln \int \exp\{\zeta'(x)\alpha_t\} dx. \quad (8)$$

Here, $\bar{\zeta}'(X_t)$ is the K -dimensional vector of sufficient statistics, and $\varphi(\alpha_t)$ is the log normalization constant for the density $p_t^{(K)}(x)$. Below, we sometimes use β as a generic parameter for the log-likelihood function $\mathcal{L}^{(K)}(\cdot|X_t)$.

The assumption that x_{it} is i.i.d. (independently and identically distributed) across i and t is consistent with datasets that comprise repeated cross sections. In practice, the assumption may not be strictly satisfied, for two reasons. First, the micro data could exhibit spatial correlation. Spatial correlation caused by a common factor can be mitigated by transforming the micro-level observations. For instance, in our application we divide earnings by the labor share of aggregate GDP, which removes comovements generated by the aggregate business cycle. If there remains some residual spatial correlation after the transformation, then our proposed estimation approach essentially replaces the likelihood function for $x_{1t}, \dots, x_{Nt}$ with a composite likelihood function that ignores the spatial correlation; see Varin, Reid, and Firth (2011).

Second, if the cross-sectional densities $p_t^{(K)}(x)$ are estimated from a panel dataset or a rotating panel, there will be some temporal dependence in the micro data that is not captured by the likelihood function. While a careful analysis is beyond the scope of this paper, we expect the large sample limits of our estimators to be unaffected because inference on the α_t 's is mainly based on cross-sectional averaging of time t observations. At the same time, there is some potential loss of information in the use of what then would be a quasi likelihood instead of the exact likelihood function. Moreover, credible intervals derived from the quasi posterior might understate "true" posterior uncertainty. However, on the positive side, the functional modeling approach does not require the econometrician to make assumptions about the time-series evolution of x_{it} for each unit i , which may exhibit complicated nonlinear dynamics, and the cross-sectional coefficient heterogeneity in the x_{it} law of motion.

To summarize, our K -dimensional functional model has a state-space representation with state vector α_t , measurement equation (8), and state-transition equation (6). When estimating the functional model, we allow for $p > 1$ lags in the state-transition equation. However, to keep the notation simple, we proceed for now with a single lag, $p = 1$.

C. Approximate Filtering

The likelihood function of the functional model can be evaluated by using a filter. To simplify the notation, we collect the parameters $(\alpha_*, Y_*, \Phi, \Sigma)$ in

the vector θ . Here the matrix Φ comprises Φ_{yy} , $\Phi_{y\alpha}$, $\Phi_{\alpha y}$, and $\Phi_{\alpha\alpha}$. Although the dimension of θ depends on the degree of approximation K , we omit the (K) superscript. Let $Y_{1:t}$ denote the sequence $Y_1, \dots, Y_t$. Starting from a distribution $p^{(K)}(\alpha_{t-1} | Y_{1:t-1}, X_{1:t-1}, \theta)$, for each period t the filter computes

$$\begin{aligned} p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) &= \int p_G^{(K)}(Y_t, \alpha_t | Y_{t-1}, \alpha_{t-1}, \theta) p^{(K)}(\alpha_{t-1} | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_{t-1}, \\ p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta) &= \int p^{(K)}(X_t | \alpha_t) p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_t, \\ p^{(K)}(\alpha_t | Y_{1:t}, X_{1:t}, \theta) &\propto p^{(K)}(X_t | \alpha_t) p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta). \end{aligned} \quad (9)$$

The first equation in display (9) iterates the state-transition equation (6) forward and integrates over the hidden state α_{t-1} . We use the subscript “G” to indicate that the state-transition equation is linear and Gaussian. The second equation generates a forecast of the observables (Y_t, X_t) using the measurement equation (8). The third equation updates the density of the hidden state α_t , using Bayes’s theorem (here, “ $\propto$ ” denotes proportionality). The density $p^{(K)}(X_t | \alpha_t)$ defined in equation (8) does not depend on θ .

Because the measurement equation for X_t is nonlinear and non-Gaussian, the predictive density $p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta)$ cannot be calculated analytically. While one could use a sequential Monte Carlo filter to generate a numerical approximation, we approximate the measurement equation $p^{(K)}(X_t | \alpha_t)$ in a way that enables analytical calculations using the Kalman filter (KF) recursions. A second-order Taylor series expansion of $\mathcal{L}(\alpha_t | X_t)$ in equation (8) around the maximum likelihood estimator

$$\hat{\alpha}_t = \operatorname{argmax}_{\beta \in \mathcal{A}} \mathcal{L}^{(K)}(\beta | X_t) \quad (10)$$

yields ($\mathcal{A}$ is defined in assumption 1 below)

$$p^{(K)}(X_t | \alpha_t) = \exp \left\{ N \mathcal{L}^{(K)}(\hat{\alpha}_t | X_t) - \frac{N}{2} (\alpha_t - \hat{\alpha}_t)' \hat{V}_t^{-1} (\alpha_t - \hat{\alpha}_t) + N \mathcal{R}(\alpha_t) \right\}, \quad (11)$$

where $\hat{V}_t$ is the negative inverse Hessian associated with $\mathcal{L}(\beta | X_t)$ evaluated at $\hat{\alpha}_t$, and $\mathcal{R}(\alpha_t)$ is the remainder term from the second-order Taylor series approximation. Define

$$\begin{aligned} p_{\text{KF}}^{(K)}(X_t | \alpha_t) &= \exp \left\{ N \mathcal{L}^{(K)}(\hat{\alpha}_t | X_t) - \frac{N}{2} (\alpha_t - \hat{\alpha}_t)' \hat{V}_t^{-1} (\alpha_t - \hat{\alpha}_t) \right\} \\ &= p_{\text{pen}}^{(K)}(X_t | \hat{\alpha}_t) \times p_G^{(K)}(\hat{\alpha}_t | \alpha_t), \end{aligned} \quad (12)$$

where

$$p_{\text{pen}}^{(K)}(X_t|\hat{\alpha}_t) = \exp\{N\mathcal{L}^{(K)}(\hat{\alpha}_t|X_t)\} \left(\frac{2\pi}{N}\right)^{K/2} |\hat{V}_t|^{1/2}.$$

We use the subscript “pen” to indicate that the expression can be interpreted as the maximized likelihood function of X_t penalized by $(2\pi/N)^{K/2}|\hat{V}_t|^{1/2}$. The $p_G^{(K)}(\hat{\alpha}_t|\alpha_t)$ term in equation (12) is the Gaussian density corresponding to the “measurement” equation⁷

$$\hat{\alpha}_t = \alpha_t + N^{-1/2}\eta_t, \quad \eta_t \sim \mathcal{N}(0, \hat{V}_t). \quad (13)$$

With the exact measurement equation $p^{(K)}(X_t|\alpha_t)$ replaced by the approximation $p_{\text{KF}}^{(K)}(X_t|\alpha_t)$, the filter iterations in equation (9) become

$$\begin{aligned} p_{\text{KF}}^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) &= \int p_G^{(K)}(Y_t, \alpha_t | Y_{t-1}, \alpha_{t-1}, \theta) p_{\text{KF}}^{(K)}(\alpha_{t-1} | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_{t-1}, \\ p_{\text{KF}}^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta) &= p_{\text{pen}}^{(K)}(X_t|\hat{\alpha}_t) \int p_G^{(K)}(\hat{\alpha}_t | \alpha_t) p_{\text{KF}}^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_t \\ &= p_{\text{pen}}^{(K)}(X_t|\hat{\alpha}_t) p_{\text{KF}}^{(K)}(Y_t, \hat{\alpha}_t | Y_{1:t-1}, X_{1:t-1}, \theta), \\ p_{\text{KF}}^{(K)}(\alpha_t | Y_{1:t}, X_{1:t}, \theta) &\propto p_G^{(K)}(\hat{\alpha}_t|\alpha_t) p_{\text{KF}}^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta). \end{aligned} \quad (14)$$

Because the state-transition density $p_G^{(K)}(Y_t, \alpha_t | Y_{t-1}, \alpha_{t-1}, \theta)$ is normal under the assumption that the shock vector u_t is normally distributed, all the densities in equation (14) are Gaussian—we initialize the filter with a Gaussian density for α_0 —and the KF recursions can be used to track means and covariance matrices. It can be verified by induction that $X_{1:t-1}$ affects the densities only through $\hat{\alpha}_{1:t-1}$. Thus, we can replace $X_{1:t-1}$ with $\hat{\alpha}_{1:t-1}$ in the conditioning sets in equation (14).

To summarize, our approximation of the filtering problem leads to a convenient two-step procedure. In the first step, the researcher computes the sequence of MLEs $\hat{\alpha}_t$ from the cross-sectional observations X_t , separately for each period $t = 1, \dots, T$. In the second step, the researcher estimates a linear state-space model in which the $\hat{\alpha}_t$'s are interpreted as noisy measures of the latent α_t 's.

D. Bayesian Model-Selection Criterion

In empirical work, the likelihood function is combined with a prior distribution $p^{(K)}(\theta|\lambda)$, which depends on a vector λ of hyperparameters that

⁷ This is not a statement about the asymptotic sampling distribution of the MLE $\hat{\alpha}_t$.

control the variance of the prior. We use the Bayesian MDD to determine λ , the sieve order K , and the number of lags in the state-transition equation (2).⁸ The MDD is defined as

$$p^{(K)}(Y_{1:T}, X_{1:T} | \lambda) = \int \left(\prod_{t=1}^T p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta) \right) p^{(K)}(\theta | \lambda) d\theta. \quad (15)$$

This expression is very difficult to evaluate numerically. As we discussed in section II.C, the computation of $p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta)$ requires a nonlinear filter. Moreover, the high-dimensional parameter vector θ has to be integrated out numerically.

We simplify the problem in two steps. First, we replace the conditional density of (Y_t, X_t) with the KF approximation and define

$$p_{\text{KF}}^{(K)}(Y_{1:T}, X_{1:T} | \lambda) = \left(\prod_{t=1}^T p_{\text{pen}}^{(K)}(X_t | \hat{\alpha}_t) \right) \int \left(\prod_{t=1}^T p_{\text{KF}}^{(K)}(Y_t, \hat{\alpha}_t | Y_{1:t-1}, \hat{\alpha}_{1:t-1}, \theta) \right) p^{(K)}(\theta | \lambda) d\theta. \quad (16)$$

Second, we set the measurement-error variance in equation (13) to zero, which will allow us to replace the KF density $p_{\text{KF}}^{(K)}(Y_t, \hat{\alpha}_t | Y_{1:t-1}, \hat{\alpha}_{1:t-1}, \theta)$ with the Gaussian state-transition density $p_G^{(K)}(Y_t, \alpha_t = \hat{\alpha}_t | Y_{1:t-1}, \alpha_{1:t-1} = \hat{\alpha}_{1:t-1}, \theta)$, so that we obtain the following MDD approximation:

$$p_*^{(K)}(Y_{1:T}, X_{1:T} | \lambda) = \left(\prod_{t=1}^T p_{\text{pen}}^{(K)}(X_t | \hat{\alpha}_t) \right) \int \left(\prod_{t=1}^T p_G^{(K)}(Y_t, \alpha_t = \hat{\alpha}_t | Y_{1:t-1}, \alpha_{1:t-1} = \hat{\alpha}_{1:t-1}, \theta) \right) p^{(K)}(\theta | \lambda) d\theta. \quad (17)$$

The integral on the right-hand side of the equation is the MDD associated with the VAR in equation (6), where the latent α 's are replaced by the MLEs $\hat{\alpha}_t$. There exists a large literature on how to evaluate VAR MDDs, and we choose a conjugate prior distribution $p^{(K)}(\theta | \lambda)$ that allows us to compute $p_*^{(K)}(Y_{1:T}, X_{1:T} | \lambda)$ analytically.

III. Approximation-Error Bounds and Model Selection

In this section, we formally characterize the accuracy of the KF and MDD approximations. We begin in section III.A by defining some notation and

⁸ MDD-based hyperparameter selection or averaging has been widely used in the VAR literature; see, for instance, Del Negro and Schorfheide (2004) and Giannone, Lenza, and Primiceri (2015).

stating a basic assumption and a lemma. We present an approximation-error bound for the KF state distribution and the likelihood increment in section III.B and provide a bound for the MDD approximation in section III.C. Model selection is discussed in section III.D. Let $f(N, K; \omega)$ be a function that depends on the sample size N , the approximation order K , and a random element ω that lives on a probability space equipped with a probability measure $\mathbb{P}^\omega$. Throughout this section, we write $|f(N, K; \omega)| \lesssim r_{K,N} = o(1)$ to mean that there is a constant C such that for every small $\epsilon > 0$ we can find (K_0, N_0) such that $\mathbb{P}^\omega\{|f(N, K; \omega)| \leq Cr_{K,N}\} \geq 1 - \epsilon$ for $K > K_0$ and $N > N_0$. Moreover, we write $f(N, K; \omega) \asymp 1 \pm r_{K,N}$ to denote that $|f(N, K; \omega) - 1| \lesssim r_{K,N}$.

A. Preliminaries: Cross-Sectional Densities

The derivatives of $\mathcal{L}^{(K)}(\beta|X)$ with respect to β of order 2 and higher depend only on derivatives of $\varphi(\beta)$; see equation (8). Let $\mathcal{L}^{(K)(2)}(\beta)$ be the matrix of second derivatives and $\mathcal{L}_{hklm}^{(K)(4)}(\beta)$ be the fourth derivatives, where the subscript indicates the elements of β with respect to which we are differentiating.

ASSUMPTION 1. The fourth derivatives $\mathcal{L}_{hklm}^{(K)(4)}(\beta)$ are uniformly bounded by a constant M for $\beta \in \mathcal{A} \subseteq \mathbb{R}^K$, where $\mathcal{A} = \{\beta \in \mathbb{R}^K : \varphi(\beta) \leq C < \infty\}$. The constant does not depend on K or X . Moreover, the matrix $[-\mathcal{L}^{(K)(2)}(\beta)]$ is positive definite at $\arg\max_{\beta \in \mathcal{A}} \mathcal{L}^{(K)}(\beta|X_t)$.

In our application, we use cubic splines and assume that $x \in [\underline{x}, \bar{x}]$. Thus, assumption 1 is satisfied. To state the following lemma, we add a (K) superscript to the remainder $\mathcal{R}(\cdot)$ and the densities $p(X_t|\cdot)$ and a $1:N$ subscript to X_t to highlight the dependence of these objects on (K, N) .

LEMMA 1. Suppose that assumption 1 is satisfied. Then, there is a finite positive constant C and an $N_0(C)$ such that uniformly for all $\alpha_t \in \text{int}(\mathcal{A})$, for all $t = 1, \dots$, and $N \geq N_0 > K^3$,

$$\exp\{N\mathcal{R}^{(K)}(\alpha_t)\} \asymp 1 \pm \sqrt{\frac{K^3}{N}}, \quad \frac{p^{(K)}(X_{1:N,t}|\alpha_t)}{p_{\text{KF}}^{(K)}(X_{1:N,t}|\alpha_t)} \asymp 1 \pm \sqrt{\frac{K^3}{N}}.$$

B. KF Approximation Error

Our first set of approximation results is about the relationship between the density of the latent state α_t , given time t information under the exact filter in equation (9) and the KF in equation (14), and the likelihood increments under the two filters. Throughout the remainder of this section, we assume that $K^3/N = o(1)$.

THEOREM 1. Suppose that assumption 1 is satisfied. Then, there exists a constant C and an $N_0(C)$ that do not depend on α_t or θ such that for all $N \geq N_0 \geq K^3$ and $t = 1, \dots, T$,

$$\frac{p^{(K)}(\alpha_t | Y_{1:t}, X_{1:N,1:t}, \theta)}{p_{\text{KF}}^{(K)}(\alpha_t | Y_{1:t}, X_{1:N,1:t}, \theta)} \asymp 1 \pm \sqrt{\frac{K^3}{N}}$$

and

$$\frac{p^{(K)}(Y_t, X_{1:N,t} | Y_{1:t-1}, X_{1:N,1:t-1}, \theta)}{p_{\text{KF}}^{(K)}(Y_t, X_{1:N,t} | Y_{1:t-1}, X_{1:N,1:t-1}, \theta)} \asymp 1 \pm \sqrt{\frac{K^3}{N}}.$$

A proof of the theorem is provided in the appendix. To provide a numerical illustration, we generate a dataset $(Y_{1:T}, X_{1:N,1:T})$ from the calibrated HA model of section V and evaluate the likelihood function at the posterior mean estimate θ_* of the functional model. The exact likelihood increments $p^{(K)}(Y_t, X_{1:N,t} | Y_{1:t-1}, X_{1:N,1:t-1}, \theta)$ are computed with a particle filter (PF), which was first developed by Gordon, Salmond, and Smith (1993). A PF approximates the distribution of $\alpha_t | (Y_{1:t}, X_{1:N,1:t}, \theta)$ through a swarm of particles $\{\alpha_t^j, W_t^j\}_{j=1}^M$. The approximation is stochastic, and its accuracy depends crucially on the mutation of the particle values between periods $t-1$ and t . We use the linearized measurement equation (13) to derive an approximately conditionally optimal proposal density for the α_{t-1}^j particle values. The general implementation of the PF follows Herbst and Schorfheide (2015), and details are provided in the appendix. Conditional on θ_* , we compute PF approximations of

$$\ln p^{(K)}(Y_{1:T}, X_{1:N,1:T} | \theta_*) = \sum_{t=1}^T \ln p^{(K)}(Y_t, X_{1:N,t} | Y_{1:t-1}, X_{1:N,1:t-1}, \theta_*)$$

for $K = 8$ and $T = 25$ and two choices of N . We use $M = 5,000$ particles. We run the stochastic filter $n_{\text{sim}} = 100$ and plot kernel density estimates for the n_{sim} log-likelihood values.

Results are plotted in figure 1. Because the PF likelihood approximations are unbiased, we first calculate the mean of the likelihood values across the n_{sim} runs of the filter and regard the log of the mean as the “exact” log-likelihood value. To make results comparable for $N = 1,250$ and $N = 10,000$, we compute $\ln p^{(K)}(Y_{1:T}, X_{1:N,1:T} | \theta_*) / (N + 2)$, where $n_y = 2$ is the number of aggregate series. The log-likelihood values shown in the figure are in deviations of the “exact” values, which are -0.3467 for $N = 1,250$ and -0.3418 for $N = 10,000$. The KF log likelihood is represented by a vertical line and the distribution of PF log-likelihood approximations by a density. Deviations of the KF value from 0 can be interpreted as approximation errors of the linear filter. A comparison between figures 1A

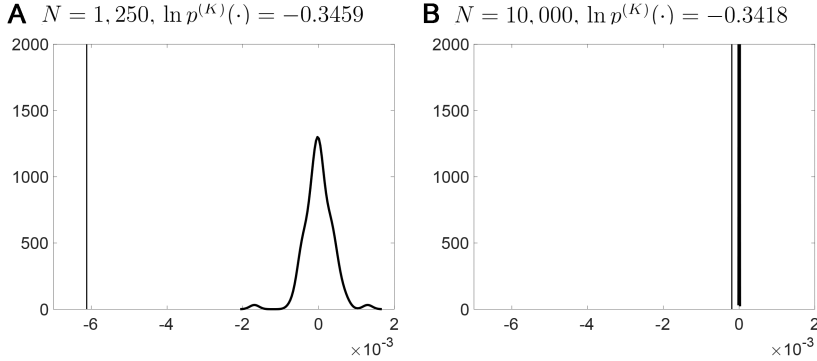


FIG. 1.—KF log-likelihood approximation errors. Log-likelihood values $\ln p^{(K)}(Y_{1:T}, X_{1:N,1:T} | \theta_*) / (N + 2)$ in deviations from the “exact” value. Vertical lines correspond to KF approximations, and densities represent distribution of PF log-likelihood values across multiple runs for $M = 5,000$ particles.

and 1B indicates that the approximation error vanishes as N increases, as theorem 1 predicts.⁹

C. MDD Approximation Error

We propose to use the MDD approximation $p_*^{(K)}(Y_{1:T}, X_{1:N,1:T} | \lambda)$ in equation (17) to select the number of lags $p \leq \bar{p}$, the sieve dimension $K \in \mathcal{K}_N$, and the prior hyperparameters $\lambda \in \Lambda$. We provide a bound for the discrepancy between the $p_*^{(K)}(\cdot)$ and the $p^{(K)}(\cdot)$ MDDs based on the nonlinear measurement equation, defined in equation (15). Because the MDD-based lag-length selection under the assumption that the upper bound $\bar{p}$ does not depend on the sample size is standard, we focus on the (K, λ) selection in the remainder of this section. We assume that the set $\mathcal{K}_N$ from which K is selected does not grow too fast. Let $\bar{K}_N = \max \{K | K \in \mathcal{K}_N\}$ and $\bar{K}_N = o(N^{1/3})$.

THEOREM 2. Suppose that assumption 1 is satisfied and that $T\bar{K}_N/N = o(1)$. Then,

$$\max_{(K, \lambda) \in \mathcal{K}_N \times \Lambda} |\ln p^{(K)}(Y_{1:T}, X_{1:N,1:T} | \lambda) - \ln p_*^{(K)}(Y_{1:T}, X_{1:N,1:T} | \lambda)| \lesssim \left(\sqrt{\frac{\bar{K}_N^3}{N}} + \frac{T}{2N} \bar{K}_N \right).$$

⁹ The dispersion of PF log-likelihood values is larger for $N = 1,250$ than for $N = 10,000$ because θ_* is the posterior mean based on the $N = 10,000$ sample and the precision of the PF is sensitive to the fit of the model/parameter values; e.g., see Herbst and Schorfheide (2015).

A proof of the theorem is provided in the appendix. The rate $(\bar{K}_N^3/N)^{1/2}$ reflects the discrepancy between $\ln p^{(K)}(\cdot)$ and $\ln p_{\text{KF}}^{(K)}(\cdot)$. The second rate, $T\bar{K}_N/N$, comes from replacing $\ln p_{\text{KF}}^{(K)}(\cdot)$ with $\ln p_*^{(K)}(\cdot)$.¹⁰

D. Model Selection

Let $(\hat{K}, \hat{\lambda})$ and (K_*, λ_*) be the maximizers of $\ln p^{(K)}(\cdot | \lambda)$ and $\ln p_*^{(K)}(\cdot | \lambda)$, respectively. We can deduce the following corollary from theorem 2:

COROLLARY 1. Suppose that assumption 1 is satisfied and that $T\bar{K}_N/N = o(1)$. Then,

$$\begin{aligned} \ln p^{(\hat{K})}(Y_{1:T}, X_{1:N,1:T} | \hat{\lambda}) - \ln p^{(K_*)}(Y_{1:T}, X_{1:N,1:T} | \lambda_*) &\lesssim 2 \left(\sqrt{\frac{\bar{K}_N^3}{N}} + \frac{T}{2N} \bar{K}_N \right), \\ \ln p_*^{(K_*)}(Y_{1:T}, X_{1:N,1:T} | \lambda_*) - \ln p_*^{(\hat{K})}(Y_{1:T}, X_{1:N,1:T} | \hat{\lambda}) &\lesssim 2 \left(\sqrt{\frac{\bar{K}_N^3}{N}} + \frac{T}{2N} \bar{K}_N \right). \end{aligned}$$

The corollary establishes a weak form of equivalence between model selection based on the exact MDD $\ln p^{(K)}(\cdot | \lambda)$ and the approximation $\ln p_*^{(K)}(\cdot | \lambda)$: the argmax of $\ln p_*^{(K)}(\cdot | \lambda)$, (K_*, λ_*) , almost maximizes the exact MDD, and vice versa.¹¹ We show in the appendix that, under a version of the drifting prior in equation (18) below and a log-likelihood function that is quadratic in θ , the hyperparameter estimates based on the exact and approximate MDDs are within distance ϵ for each $K \in \mathcal{K}_N$ as $(N, T) \rightarrow \infty$. A strong equivalence statement regarding the K selection requires assumptions on a DGP $\mathbb{P}_{N,T}^0(\cdot)$, and the notion of a pseudo-optimal dimension $K_{oN} \in \mathcal{K}_N$. A proof is left for future research.

To understand the trade-offs in choosing λ and K , we now decompose $\ln p_*^{(K)}(\cdot)$ into goodness-of-fit and penalty terms. For ease of exposition, we assume that the log-likelihood function and prior density are both quadratic in θ . In our model, this assumption is exactly satisfied for the regression coefficients Φ and holds asymptotically for large T for the parameters in Σ . This assumption avoids having to keep track of standard remainder terms. To keep the hyperparameter selection asymptotically nontrivial,

¹⁰ In the empirical application in sec. VI, N varies with t . The minimum value is 12,411. The time-series length is $T = 110$. Most of the calculations are based on $K = 10$, which leads to $K^3/N = 0.08$, $TK/(2N) = 0.06$. For MDD comparisons, we consider $\bar{K} = 22$, which leads to $\bar{K}^3/N = 0.86$ and $T\bar{K}/(2N) = 0.10$. Because the law of motion for α_t induces a prior for the spline coefficient, we allowed $\bar{K}$ to be slightly larger than the rate restriction suggests.

¹¹ In the context of selecting the dimension of a linear model, Shao (1997) refers to the weak notion as loss efficiency. He compares the behavior of various model-selection procedures under finite-dimensional and infinite-dimensional data-generating processes (DGPs).

we balance the informational contents of the prior distribution and the likelihood function by rescaling the hyperparameter. We assume that

$$\theta|(K, \omega) \sim \mathcal{N}\left(\underline{\theta}, \frac{\omega}{T} \underline{\mathcal{V}}\right). \quad (18)$$

The local hyperparameter $\omega \in \Omega = [0, \infty)$ is related to the precision hyperparameter λ through $\lambda = T/\omega$.

Rather than conducting a Laplace approximation around the MLE $\hat{\theta}_*$, we first compute the posterior mean/mode and variance to capture the effect of the prior distribution in the MDD formula. In fact, under the assumption of a log-quadratic likelihood and prior density, the following formulas are exact. The posterior of Θ is Gaussian, with mean and variance (scaled by T)

$$\bar{\theta}_* = \bar{\mathcal{V}}_* \left(\hat{\mathcal{V}}_*^{-1} \hat{\theta}_* + \frac{1}{\omega} \underline{\mathcal{V}}^{-1} \underline{\theta} \right) \quad \text{and} \quad \bar{\mathcal{V}}_* = \left(\hat{\mathcal{V}}_*^{-1} + \frac{1}{\omega} \underline{\mathcal{V}}^{-1} \right)^{-1},$$

where $\hat{\mathcal{V}}_*$ is the negative inverse Hessian associated with $\ln p_G^{(K)}(Y_t, \alpha_t = \hat{\alpha}_t | \cdot, \theta)$, evaluated at the MLE $\hat{\theta}_*$. This leads to, say,

$$\begin{aligned} & \frac{1}{NT} \ln p_*^{(K)}(Y_{1:T}, X_{1:N,1:T} | \omega) \\ &= \frac{1}{T} \sum_{t=1}^T \mathcal{L}^{(K)}(\hat{\alpha}_t | X_{1:N,t}) - \frac{K}{2N} \ln(N) \\ & \quad - \frac{1}{2N} \left(\hat{\theta}_*' \hat{\mathcal{V}}_*^{-1} \hat{\theta}_* + \frac{1}{\omega} \underline{\theta}' \underline{\mathcal{V}}^{-1} \underline{\theta} - \bar{\theta}_*' \bar{\mathcal{V}}_*^{-1} \bar{\theta}_* \right) - \frac{1}{2NT} \ln |\omega \underline{\mathcal{V}} \hat{\mathcal{V}}_*^{-1} + I_d| \\ & \quad + \frac{1}{NT} \sum_{t=1}^T \ln p_G^{(K)}(Y_t, \alpha_t = \hat{\alpha}_t | Y_{t-1}, \alpha_{t-1} = \hat{\alpha}_{t-1}, \hat{\theta}_*) \\ & \quad + \frac{K}{2N} \ln(2\pi) + \frac{1}{2} \frac{1}{NT} \sum_{t=1}^T \ln |\hat{V}_t| \\ &= (A_1 - A_2) + (-A_3 - A_4) + (A_5 + A_6 + A_7). \end{aligned} \quad (19)$$

Note that the dimension of θ is $O(K^2)$.

The $O_p(1)$ term $A_1 = (1/T) \sum_{t=1}^T \mathcal{L}^{(K)}(\hat{\alpha}_t | X_{1:N,t})$ captures the goodness of fit of the cross-sectional density estimates and is nondecreasing in K . It is penalized by the Schwarz criterion (BIC [Bayesian information criterion]) penalty $A_2 = K \ln N / (2N)$, meaning that every additional sieve term is penalized by $\ln N / (2N)$.¹² The selection of ω conditional on K

¹² In fact, Kooperberg and Stone (1992) recommend K selection by BIC for log-spline density models.

depends on A_3 and A_4 .¹³ For finite values of ω , the term A_3 is greater than zero and can be interpreted as an in-sample goodness-of-fit adjustment to the maximized VAR likelihood in A_5 , because of the use of a prior. As $\omega \rightarrow \infty$, meaning that the prior becomes “flat,” the difference between the posterior mean and the MLE vanishes and A_3 converges to zero. The term A_4 penalizes the choice of the hyperparameter ω . The penalty diverges as $\omega \rightarrow \infty$. Also, A_4 is minimized for $\omega = 0$ when θ is forced to be equal to the prior mean $\underline{\theta}$. The remaining terms, A_5 , A_6 , and A_7 , are of the smaller order $O_p(K/N)$.

To summarize, dimensionality and hyperparameter selection based on the approximate MDD $p_*^{(K)}(Y_{1:T}, X_{1:N,1:T}|\lambda)$ trades off goodness of fit and model complexity captured by the penalty terms A_2 and A_4 . By maximizing the approximate MDD, the user can speed up computations considerably, because it simplifies a high-dimensional integration problem. In our application, we have an analytical formula for the $p_*^{(K)}(\cdot|\lambda)$ MDD. The $p^{(K)}(\cdot|\lambda)$ MDD value attained by this computational strategy is almost as high, in the sense of corollary 1, as that obtained by maximizing the exact MDD, $p_*^{(K)}(Y_{1:T}, X_{1:N,1:T}|\lambda)$.

IV. Implementation Details

Basis functions.—A convenient basis for the log density is a spline that is a piecewise polynomial function with knots x_k , $k = 1, \dots, K - 1$; see Kooperberg and Stone (1991). A typical choice is to consider a cubic spline that is restricted to be linear and upward sloping on the interval $(-\infty, x_1)$ and linear and downward sloping on the interval $[x_{K-1}, \infty)$. Thus, the estimated density has the tails of a Laplace density, which are a bit thicker than Gaussian tails.¹⁴ In our two applications, we restrict x to the interval $[0, \bar{x}]$. For the estimation based on simulated data in section V, we also use a cubic function for the last segment of the spline:

$$\begin{aligned}\zeta_1(x) &= x, \\ \zeta_k(x) &= [\max\{x - x_{k-1}, 0\}]^3, \quad k = 2, \dots, K.\end{aligned}\tag{20}$$

We exclude the constant function $\zeta_0(x) = 1$ because it is redundant in light of the normalization imposed in the definition of $\mathcal{L}^K(\alpha_t|X_t)$ in equation (8). For the empirical application in section VI, we construct the

¹³ The term A_4 is $O_p(K^2/(NT))$. The term A_5 is also of order $O_p(K^2/(NT))$ if two conditions hold. First, $K^2/T = o(1)$, which delivers a $\sqrt{T}$ -consistent MLE. Second, the “true” θ parameter is sampled from the drifting prior (eq. [18]), which implies that the MLE behaves asymptotically as $\hat{\theta}_s = \underline{\theta} + \hat{\theta}_s/\sqrt{T}$, where $\hat{\theta}_s = O_p(1)$.

¹⁴ In general, the methods developed in this paper are not restricted to cubic splines. Other basis functions, such as polynomials (including trigonometric, Hermite, Laguerre) or wavelets, could be used.

spline from $x = \bar{x}$ to $x = 0$, rather than from $x = 0$ to $x = \bar{x}$, using a linear element for the right tail:

$$\begin{aligned}\zeta_K(x) &= \max\{\bar{x} - x, 0\}, \\ \zeta_k(x) &= [\max\{x_{k-1} - x, 0\}]^3, \quad k = K - 1, \dots, 1.\end{aligned}\tag{21}$$

For the simulated data, equation (20) is preferred because the right tail of the simulated asset distributions is very thin. For the actual data, it is desirable that the rightmost segment of the spline is linear, because under the cubic specification the density is increasing between the top-coded value (see below) and the upper bound $\bar{x}$ in some periods. Near zero, on the other hand, the cubic segment is more successful approximating the density than the linear segment. As a diagnostic, we recommend visually examining the fit of the estimated cross-sectional densities, as in figures 4 and 8. In addition, the MDD $p_*^{(K)}(Y_{1:T}, X_{1:T}|\lambda)$ can be used to select among basis functions, which is what we did to choose between equations (20) and (21) in the simulation experiment of section V and the empirical analysis in section VI.

Sieve coefficients.—The first step in the estimation of the functional model is the computation of the MLEs $\hat{\alpha}_t$ for $t = 1, \dots, T$. In this step, we also generate the sequence of covariance matrices $\hat{V}_t$, and the MDD components $p_{\text{KF}}^{(K)}(X_t|\alpha_t)$, $t = 1, \dots, T$. The appendix discusses three adjustments that may be required to implement the empirical analysis: (i) If the data are top-coded, the likelihood must be adjusted. We assume that the censoring point and the number of censored observations are observed, and $P_t(x)$ is assumed to describe the uncensored density. (ii) Even though K basis functions may be necessary to approximate the cross-sectional densities, the time variation might be concentrated in a lower-dimensional space, because, for instance, only the means of the cross-sectional distributions are varying over time. If there are perfect collinearities in $\hat{\alpha}_t$, we remove them. Even if there are none, we normalize the $\hat{\alpha}_t$'s. (iii) The cross-sectional observations may exhibit seasonality that leads to seasonal variation in the $\hat{\alpha}_t$'s. In our application in section VI, the aggregate data are seasonally adjusted. Thus, we remove deterministic seasonalities from the $\hat{\alpha}_t$'s before combining them with the aggregate variables in a VAR.

Priors and posteriors.—The second step of the analysis consists of the estimation of the state-space model, based on the sequence $(\hat{\alpha}_t, \hat{V}_t)$, $t = 1, \dots, T$, generated in the first step. In our implementation, we condition on estimates of the deterministic components, $\hat{Y}_*$ and $\hat{\alpha}_*$. Recall that the measurement equation is given by equation (13) and that we assume that $Y_t - Y_*$ is observed without error. Define the $n_w \times 1$ vector $W_t = [(Y_t - Y_*)', (\alpha_t - \alpha_*)']'$, and write the multilag generalization of state-transition equation (6) more compactly as

$$W_t = \sum_{h=1}^p \Phi_h W_{t-h} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma). \quad (22)$$

We now also allow for multiple lags. Define $\Phi = [\Phi_1, \dots, \Phi_p]'$, and note that (Φ, Σ) do not enter the measurement equation.

Following Carter and Kohn (1994), posterior inference for the state-space model is based on a Gibbs sampler that iterates over the following two conditional posterior distributions:

$$\alpha_{1:T} | (Y_{1:T}, X_{1:T}, \Phi, \Sigma) \quad \text{and} \quad (\Phi, \Sigma) | (Y_{1:T}, X_{1:T}, \alpha_{1:T}).$$

Because we are using the linearized measurement equation (13), the latent states $\alpha_{1:T}$ can be sampled with a KF/simulation smoother. We adjust the standard algorithm to account for the aggregate variables Y_t being directly observed and multiple lags, that is, $p > 1$. Details are provided in the appendix. Conditional on $\alpha_{1:T}$, the derivation of the posterior distribution of (Φ, Σ) follows the Bayesian analysis of a VAR. Because the vector α_t can be large, we use the VAR parameterization and prior proposed by Chan (2022). His specification is suitable for high-dimensional settings because it leads to equation-by-equation estimation while allowing for some asymmetry of the prior across equation.

The starting point for the specification of the prior is the assumption that the elements of Φ are inverse-gamma-scale mixtures of normals with mean zero. The overall prior precision is controlled by the hyperparameter λ_1 . The hyperparameter λ_2 controls the relative precision of coefficients that capture the effect of lagged α_t on current Y_t . As $\lambda_2 \rightarrow \infty$, the cross-sectional density does not Granger-cause the aggregate variables, and the system becomes block triangular. Because Chan's (2022) prior is conjugate, the posterior distribution has a convenient inverse-gamma normal form, and draws can be generated by direct sampling. Moreover, the MDD in equation (17) can be evaluated analytically. Further details on the prior distribution, the derivation of the posterior, and the MDD formula are provided in the appendix.

Recovering cross-sectional densities.—On the basis of the estimated state-transition equation (22), we can generate forecasts and IRFs for the compressed sieve coefficients. However, the dynamics of these coefficients in themselves are not particularly interesting. Thus, we have to convert the coefficients back into densities, using

$$p^{(K)}(x | \alpha_t) = \frac{\exp\{\check{\zeta}'(x)\alpha_t\}}{\int \exp\{\check{\zeta}'(\tilde{x})\alpha_t\} d\tilde{x}}.$$

Further shortcuts.—We generally recommend estimating the functional model with the approximate state-space representation that accounts for the estimation of the α_t 's with the measurement equation (13). However,

as the calculations underlying theorem 2 indicate, the large (N, K, T) approximation-error calculations would also justify the estimation of the VAR parameters θ by simply equating the latent α_t with $\hat{\alpha}_t$, that is, setting the measurement error in equation (13) to zero and estimating the VAR parameters based on

$$\prod_{t=1}^T p_G^{(K)}(Y_t, \alpha_t = \hat{\alpha}_t | Y_{t-1}, \alpha_{t-1} = \hat{\alpha}_{t-1}, \theta), \quad (23)$$

in the notation of equation (17). In the notation of equation (22), this amounts to replacing W_t with $\hat{W}_t = [(Y_t - Y_*)', (\hat{\alpha}_t - \alpha_*)']'$. To facilitate this shortcut, we provide Julia code that (i) estimates $\hat{\alpha}_t$ based on X_t for $t = 1, \dots, T$ and (ii) converts vectors α into cross-sectional densities $p^{(K)}(x|\alpha)$. Empirical researchers can then easily combine this code with their preferred VAR-estimation routine.

Mixed-frequency considerations.—In some applications, the cross-sectional observations x_{it} might be observed at a lower frequency—say, annually instead of quarterly—than the aggregate time series. Suppose that x_{it} corresponds to household earnings and that the length of the period t is 1 quarter. It is important to distinguish two cases. Suppose that at the end of every year the household is asked, How much did you earn in the fourth quarter of the current year? In this case, we observe draws x_{it} from $p_t(x)$ in periods $t = 4, 8, 12, \dots$, and we can write equation (13) as

$$\hat{\alpha}_t = \begin{cases} \alpha_t + N^{-1/2} \eta_t & \text{for } t = 4, 8, 12, \dots, \\ \emptyset & \text{for other periods.} \end{cases}$$

The estimation can be easily handled within the current state-space setup by using the approach in Schorfheide and Song (2015). Alternatively, suppose that at the end of every year the household is asked, How much did you earn in the current year? Thus, the survey generates observations of the form $\bar{x}_{it} = x_{it} + x_{it-1} + x_{it-2} + x_{it-3}$, where now i refers to the same household in the four periods. Our model does not have enough structure to recover the distribution of $\bar{x}_{it}$ from the sequence $p_t(x), p_{t-1}(x), p_{t-2}(x), p_{t-3}(x)$. Additional assumptions would be required, and we leave a careful treatment of this case for future research.

V. A Simulation Experiment

To examine our functional state-space model's ability to capture the joint dynamics of aggregate variables and a cross-sectional distribution, we estimate it on the basis of artificial data simulated from a KS (Krusell and Smith 1998) economy. We show that in the perfect-aggregation environment of the KS model, estimated responses of aggregate variables to an aggregate shock from the functional model are identical to those

from a VAR in aggregate variables only. The MDD hyperparameter selection shrinks to zero the estimates of the parameter block that captures the effect of the lagged distribution onto current aggregates. If the DGP is modified to break the perfect-aggregation structure, some discrepancies between the two IRF estimates emerge, but considerable overlap between the credible bands remains. We also document that the estimated functional model can reproduce the distributional responses of the DGP.

A. Model Economy and DGP

The model economy consists of a continuum of households $j \in [0, 1]$. Household j chooses consumption and asset holdings to maximize

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{c_{jt}^{1-\sigma} - 1}{1-\sigma} \right],$$

subject to the budget constraint

$$x_{jt+1} + c_{jt} = (1 - \tau) W_t \epsilon_{jt} + b W_t (1 - \epsilon_{jt}) + x_{jt} (1 + R_t), \quad x_{jt+1} \geq \underline{x}. \quad (24)$$

Here, $\epsilon_{jt} \in \{0, 1\}$ is an exogenous two-state Markov process that determines the efficiency units of labor supplied by the household j in period t . Households with $\epsilon_{jt} = 1$ receive after-tax labor income $(1 - \tau) W_t$, and households with $\epsilon_{jt} = 0$ receive unemployment benefits $b W_t$. The total labor supply $L = \int \epsilon_{jt} dj$ is fixed over time. The government is assumed to balance its budget in each period by setting $\tau = b(1 - L)/L$. The asset x_{jt} is a claim on the aggregate capital stock and generates a risky return R_t . Households face the borrowing constraint $x_{jt+1} \geq \underline{x}$. Finally, a representative firm produces output Y_t according to

$$Y_t = \exp\{z_t\} K_t^\alpha L^{1-\alpha}, \quad z_t = \rho_z z_{t-1} + \sigma_z \omega_t, \quad \omega_t \sim \mathcal{N}(0, 1), \quad (25)$$

where z_t is an exogenous aggregate-productivity shock that follows an AR(1) law of motion and K_t is the aggregate capital stock. In equilibrium, $K_t = \int x_{jt} dj$.

The aggregate state of the economy is $S_t = (z_t, \mu_t)$, where μ_t is the distribution of households over (ϵ_{jt}, x_{jt}) pairs. It is convenient to use $\mu_{t,\epsilon}$ to denote the conditional distribution of assets, given the employment status ϵ_t . To solve the model, we use the approach by Winberry (2018). Details are provided in the appendix. The solution represents $\mu_{t,\epsilon}$ as a mixture of a discrete and a continuous part:

$$q_{t,\epsilon}(x) = \hat{m}_{t,\epsilon} \Delta_{\underline{x}}(x) + (1 - \hat{m}_{t,\epsilon}) p_{t,\epsilon}(x). \quad (26)$$

TABLE 1
CALIBRATION OF KRUSELL-SMITH (KS) ECONOMY

Parameter	Definition	Value	Parameter	Definition	Value
β	Discount factor	.93	b	Unemployment benefits	.15
σ	Utility curvature	1	$\pi(\epsilon_0 \rightarrow \epsilon_1)$	Unemployment to employment	.5
$\underline{x}$	Borrowing constraint	0	$\pi(\epsilon_1 \rightarrow \epsilon_0)$	Employment to unemployment	.038
α	Capital share	.36	ρ_z	TFP persistence	.859
δ	Capital depreciation	.10	σ_z	TFP innovation SD	.028

Here, $\hat{m}_{t,\epsilon}$ is the mass of individuals for whom the borrowing constraint $\underline{x}$ is binding, and $\Delta_{\underline{x}}(x)$ is the Dirac function.¹⁵ The continuous part of the asset distribution is parameterized in terms of the time-varying centralized moments $m_{t,\epsilon,k}$ and represented as

$$p_{t,\epsilon}(x) = \exp\left\{\gamma_{t,\epsilon,0} + \gamma_{t,\epsilon,1}(x - m_{t,\epsilon,1}) + \sum_{k=2}^3 \gamma_{t,\epsilon,k}[(x - m_{t,\epsilon,1})^k - m_{t,\epsilon,k}]\right\}. \quad (27)$$

The coefficients $\gamma_{t,\epsilon,k}$ are determined such that the density integrates to 1 and is consistent with the moments $m_{t,\epsilon,k}$.

We calibrate the KS economy to loosely match features of annual US data. The parameterization is summarized in table 1. The Winberry (2018) solution generates a vector autoregressive law of motion for technology z_t , capital K_t , the mass $\hat{m}_{t,\epsilon}$ of employed and unemployed households at the borrowing constraint, and the moments $m_{t,\epsilon,k}$. We simulate this law of motion for 160 periods and draw i.i.d. cross-sectional observations from equation (26), where $p_{t,\epsilon}(x)$ is defined in equation (27). Figure 2A depicts the time series of the aggregate capital stock from $t = 1$ to $t = 160$. The capital stock reaches a trough in period $t = 115$ and peaks in period $t = 155$.

We standardize household-level asset holdings by the aggregate capital stock and focus on the cross-sectional distributions of x_{it}/K_t . The densities for the unemployed, employed households, and the two types combined are plotted in figures 2B and 2C. Because the unemployment rate is only 7%, the overall asset distribution is very similar to the distribution of employed households. We condition on $x > \underline{x} = 0$ and plot $p_{t,\epsilon}(x)$, given in equation (27). The densities illustrate that employed households hold more assets than unemployed households and that in a boom period (high capital stock due to favorable technology shocks) the mode of the asset/capital distribution shifts to the right. Because of the normalization, the mean is equal to 1 in both periods.

Our version of the KS model features perfect aggregation, meaning that the VAR(1) law of motion for $(z_t, K_t, \{\hat{m}_{t,\epsilon}\}, \{m_{t,\epsilon,k}\})$ has a lower-triangular

¹⁵ The Dirac function has the property that $\Delta_{\underline{x}}(x) = 0$ for $x \neq \underline{x}$ and $\int \Delta_{\underline{x}}(x)dx = 1$.

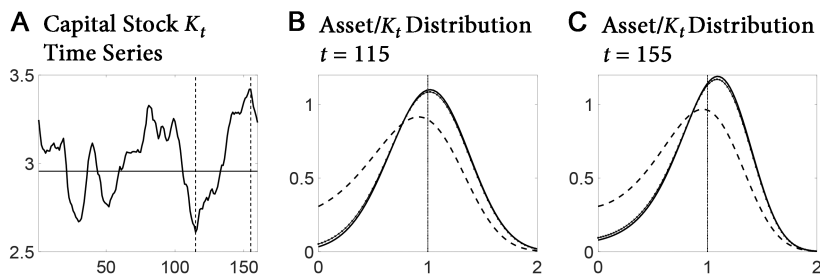


FIG. 2.—Features of simulated data (baseline). Aggregate capital attains its minimum in period $t = 115$ and its maximum in period $t = 155$. Each density is normalized so that it integrates to 1: distribution of unemployed (dashed line), distribution of employed (plain solid line), and aggregated distribution (solid line with crosses).

structure in which (z_t, K_t) does not depend on $(\{\hat{m}_{t-1,\epsilon}\}, \{m_{t-1,\epsilon,k}\})$ and thus the cross-sectional distribution does not Granger-cause the aggregate variables. In figure 3A, we compare the response of capital in the baseline KS model to a TFP shock with responses generated from a VAR(1) in the aggregate variables (z_t, K_t) , henceforth “aggregate VAR(1).”¹⁶ Under the baseline KS model, the two IRFs are identical, meaning that aggregate capital perfectly summarizes the feature of the cross-sectional asset distribution that is relevant for aggregate dynamics. In figure 3B, we consider a case in which aggregation is imperfect. Rather than specifying an alternative HA model, we simply modify the baseline KS solution by letting the lagged second moment of the asset distribution affect capital.¹⁷ Under this modification, the response of the (modified) HA model differs from the response computed from the aggregate VAR(1).

B. Functional Model Estimation

We now estimate the functional model based on the simulated time series for (z_t, K_t) and cross-sectional data.¹⁸ Unless otherwise noted, results are generated from the baseline KS model that features perfect aggregation. The econometrician observes z_t , K_t , and i.i.d. draws from the cross-sectional distribution $x_{it} \sim p_{t,\epsilon=1}(x)$, where $p_{t,\epsilon=1}(x)$ is defined in equation (27). Subsequent quantitative results are based on standardizing asset holding by the aggregate capital stock K_t . The empirical model differs from the DGP

¹⁶ Rather than explicitly solving an RA model, we simply derive a VAR(1) law of motion for (z_t, K_t) from the VAR solution of the KS model, using population OLS (ordinary least squares).

¹⁷ Kaplan and Violante (2022) survey a variety of structural mechanisms that break the perfect aggregation in the KS model.

¹⁸ As a result of precautionary savings behavior, the mass of employed individuals with zero assets is essentially zero and does not vary over time. Hence, we exclude it from the list of aggregate variables.

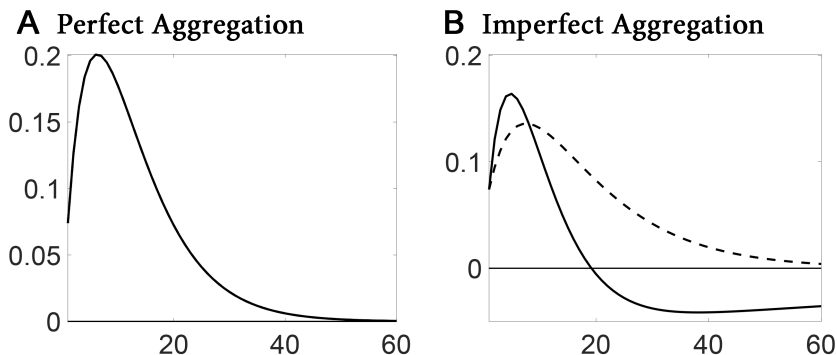


FIG. 3.—Capital stock response to TFP shock: HA model solution (solid line), population approximation in terms of aggregate variables (dashed line). A, Baseline KS model with perfect aggregation. B, Modified KS solution.

in the following important dimension: the representation of the densities in the DGP is based on the moments $m_{t,\ell,b}$ whereas the empirical model is based on a VAR that includes the spline coefficient vector α_ℓ . For a sufficiently large sieve dimension K , we expect the empirical model to deliver a good approximation to the DGP, but for any given K it does not nest the DGP. We document below that the estimated functional model is successful in capturing the joint dynamics of the cross-sectional asset distribution and the technology shock for data generated from the KS model.

1. Model Selection

The first step in the analysis is the determination of the approximation order K , the lag length p , and the hyperparameter λ based on the MDD approximation $p_*^{(K)}(Y_{1:T}, X_{1:T}|\lambda)$ in equation (17). We use the spline basis in equation (20) and place the knots at predetermined percentiles of the empirical distribution of the simulated x_{it} 's, pooled across a large simulation with $N_{\text{sim}} = 10,000$ and $T_{\text{sim}} = 160$. Table 2 summarizes the knot locations as a function of K . For $K = 4$, we use the 25th, 50th, and 75th percentiles. As K increases, we add lower and upper percentiles. Moving from $K = 8$ to $K = 10$, we add only percentiles in the left tail of the distribution, because this part of the distribution is most affected by business-cycle variations. For the subsequent estimation, we consider two choices for T , namely, $T = 90$ and $T = 150$, and three choices of N : 1,000, 5,000, and 10,000. Conditional on N , we evaluate the approximate MDD $p_*^{(K)}(Y_{1:T}, X_{1:T}|\lambda, p)$ in equation (17) over a grid of lag lengths p , sieve dimensions K , and λ values. For λ_1 and λ_2 , respectively, we consider 31 equally spaced values of $\ln \lambda_j$ between -10 and 20 .

TABLE 2
KNOT PLACEMENT

K	PERCENTILES																				
	.01	.025	.05	.1	.15	.2	.25	.3	.35	.4	.45	.5	.55	.6	.65	.7	.75	.8	.85	.9	.95
4							X					X					X				
6					X		X					X					X			X	
8			X	X			X					X					X			X	X
10			X	X	X		X					X					X			X	X
14	X	X	X	X	X		X		X			X			X		X		X	X	X
22	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

NOTE.—Percentiles (using, e.g., .01 for 1st percentile) refer to distribution of pooled observations (across $i = 1, \dots, N$ and $t = 1, \dots, T$) for $N = 10,000$ and $T = 160$.

In table 3, we report $\hat{p}$, $\hat{K}$, $\hat{\lambda}_1$, and $\hat{\lambda}_2$ for data from the baseline specification (perfect aggregation) and the modified specification (imperfect aggregation). In 11 out of 12 cases, the selected lag length is $\hat{p} = 1$, which is consistent with the first-order autoregressive structure of the DGP. For $N = 1,000$, the estimated spline order is $\hat{K} = 6$, whereas it is $\hat{K} = 8$ for $N = 5,000$ and $N = 10,000$. Thus, as the cross-sectional dimension of the sample increases, the goodness-of-fit gain in the MDD criterion through additional sieve terms outweighs the penalty. Because for any fixed K the log-spline density specification does not nest the true cross-sectional density, there is no “true” K in this simulation design.

A comparison between the baseline specification and the modified specification indicates that $\hat{\lambda}_2$, the estimated degree of relative shrinkage (to zero) for the VAR coefficients that control the effect of α_{t-1} on Y_t , $\Phi_{y\alpha}$ in equation (6), is much higher under perfect aggregation than under imperfect aggregation. This is consistent with the perfect-aggregation specification being block triangular, that is, $\Phi_{y\alpha} = 0$. The hyperparameter $\hat{\lambda}_1$ is, in 11 out of 12 cases, weakly smaller for $T = 150$ than for $T = 90$. The larger the time dimension, the smaller the need to add prior information to reduce the posterior variance.

TABLE 3
MODEL SELECTION

N	T = 90				T = 150			
	$\hat{p}$	$\hat{K}$	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{p}$	$\hat{K}$	$\hat{\lambda}_1$	$\hat{\lambda}_2$
Baseline specification: imperfect aggregation:								
1,000	1	6	7.39	148	1	6	2.72	5E−8
5,000	1	8	7.39	5E−8	1	8	7.39	5E−8
10,000	1	8	7.39	5E−8	1	8	7.39	5E−8
Alternative specification: imperfect aggregation:								
1,000	1	6	20.08	7.39	2	6	1.00	1,096
5,000	1	8	20.08	2.27	1	8	20.08	2.72
10,000	1	8	20.08	2.72	1	8	7.39	2.72

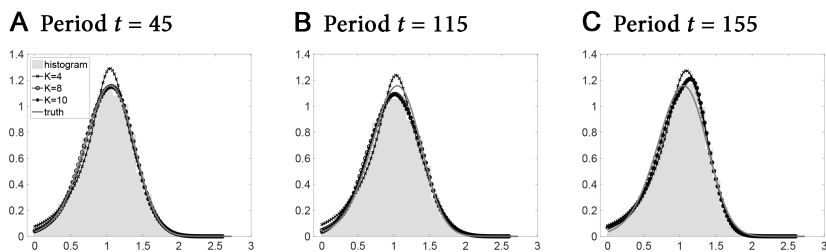


FIG. 4.—Cross-sectional fit ($N = 10,000$, $T = 150$, $K = 8$ selected). Each panel overlays a histogram of the empirical asset distribution of the employed, the log-spline density estimates for various K , and the true density from which the asset values were sampled.

2. Cross-Sectional Fit

In figure 4, we compare the fit of the estimated cross-sectional densities (various K) for $N = 10,000$,

$$p^{(K)}(x|\hat{\alpha}_t) = \exp\left\{\zeta'(x)\hat{\alpha}_t - \ln \int \exp\{\zeta'(\tilde{x})\hat{\alpha}_t\} d\tilde{x}\right\}, \quad (28)$$

to the “true” cross-sectional density and a histogram of the cross-sectional observations. The three panels correspond to different time periods. Recall from figure 2 that the capital stock reaches a trough in period $t = 115$ and peaks in period $t = 155$. The estimated densities for $K = 8$ and $K = 10$ are virtually indistinguishable and provide a smooth approximation of the histograms constructed from the observations. Because there is no improvement in fit, the MDD criterion selects $K = 8$. The $K = 4$ density shows noticeable differences from the histograms in all periods. In period $t = 45$, the density estimate for the selected $\hat{K} = 8$ almost perfectly matches the true cross-sectional density, whereas there are some small discrepancies in the modal area for the other two periods. Overall, the density estimation works well in this simulation.

3. Aggregate IRFs

We proceed by estimating the state-space representation of the functional model based on the ($N = 10,000$, $T = 150$) sample for the model specification reported in equation (3).¹⁹ In addition to the functional model, we also estimate a VAR in the aggregate variables (z_t , K_t) only, using the same prior as for the functional model. Figure 5 shows the impulse response of capital to a 3-SD (standard deviation) technology shock. We compute “true” responses from the solution of the KS economy and

¹⁹ We generate 11,000 draws from the posterior and drop the first 1,000.

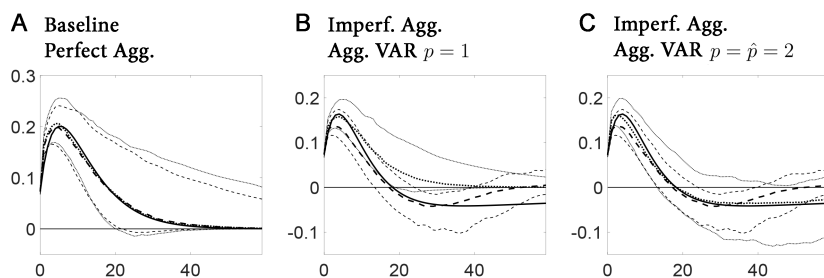


FIG. 5.—Capital response from functional model and aggregate VAR: 10th, 50th, and 90th percentiles of the posterior distribution of IRFs. Dashed bands correspond to functional VAR IRFs and dotted bands to IRFs from the aggregate VAR. Solid lines represent “true” responses from KS economy. A, Estimation based on data from baseline specification. B, C, Estimations based on data from the alternative specification with imperfect aggregation.

estimated responses (10th, 50th, and 90th percentiles of the posterior distribution) from the functional model and the aggregate VAR. Because z_t is exogenous in the KS economy, its innovation can be easily identified using a Cholesky factorization of Σ .

Figure 5A is based on data from the baseline perfect-aggregation specification. Here, the IRFs from the functional VAR(1) and the aggregate VAR(1) in (z_t, K_t) line up almost perfectly. This is expected, in view of the perfect match of the population responses in figure 3A. The cross-sectional distribution does not Granger-cause the aggregate variables, and, as a consequence, no additional lags are required in the aggregate VAR to recover the capital stock IRF.

The IRFs in figures 5B and 5C are obtained from estimates based on imperfect-aggregation data. In figure 5B we are forcing $p = 1$, which is the lag order of the functional VAR, and observe a divergence of the IRFs from functional and aggregate models. The divergence is in line with the discrepancies in figure 3B. If the cross-sectional distribution Granger-causes the aggregate variables, then (z_t, K_t) no longer evolves as a VAR(1). Instead, the bivariate process takes the form of a VARMA (VAR moving-average) process, which, strictly speaking, leads to a VAR(∞).²⁰ In finite samples, the infinite-order process has to be approximated by a finite-dimensional VAR. These considerations suggest that the lag length in the aggregate VAR must be increased to recover the capital stock IRF of the functional VAR. In figure 5C, we generate the IRFs from the aggregate model based on the number of lags $\hat{p} = 2$, estimated by the MDD. In this

²⁰ Here, the VARMA process is restricted and implies that z_t evolves according to an exogenous AR(1) process. Thus, the identification of the technology shock as z_t innovation remains unchanged.

case, functional and aggregate responses line up well. An additional lag of the aggregate variables substitutes for the missing cross-sectional information.²¹

More generally, we recommend reporting IRFs of aggregate variables from the functional model, an aggregate VAR with the same number of lags as the functional model, and possibly an aggregate VAR with data-driven lag choice. To assess HA models, one can compare the discrepancy between functional and aggregate IRFs on the basis of model-generated and actual data. If the two discrepancies differ, then the aggregation implications of the HA model are inconsistent with the data.

4. Density IRFs

For each posterior draw (Φ, Σ) we generate an IRF for α_t by iterating equation (22) forward and then reconstruct the density $p^K(x|\alpha_{t+h})$. Figure 6 shows the difference between the shocked and the steady-state densities at horizons $h = 5, 15$, and 25 .²² We refer to this difference as the response of the density. The true responses from the KS model are plotted as solid lines. At all horizons, the posterior median response aligns well with the true response, indicating that the estimated functional model can reproduce the cross-sectional dynamics of the KS model. According to the responses, probability mass shifts from $\text{asset}/K_t < 1$ to $\text{asset}/K_t > 1$ while at the same time the mass for $\text{asset}/K_t > 1.5$ decreases, keeping the mean of the cross-sectional distribution at 1.

5. Percentile IRFs

Once we have obtained the IRFs for the cross-sectional densities, we can compute IRFs for statistics derived from these densities. Here we consider percentiles of the cross-sectional distribution. We compare the percentile IRFs derived from the functional model to those obtained by estimating a VAR that combines the aggregate data with sample percentiles (10th, 20th, 50th, 80th, and 90th) computed from the cross-sectional observations. Results are plotted in figure 7. Two observations stand out. First, the estimated percentile responses from the functional model line up well with the true responses. The 10th percentile and the median of the asset/K_t distribution respond positively, whereas the 90th percentile falls. Thus, the asset share of low-wealth households increases. In turn, the asset share of

²¹ In the appendix, we report results from a second simulation experiment, based on a stylized HANK model of Auclert and Rognlie (2020). This model can generate imperfect aggregation based on countercyclical income risk. However, empirically the effect is weak. On the basis of data generated from this model, the estimated IRFs from the functional model and the aggregate VAR are very similar, even for identical lag lengths.

²² Timing convention: the system is in steady state in period $h = 1$, and the shock occurs at $h = 2$.

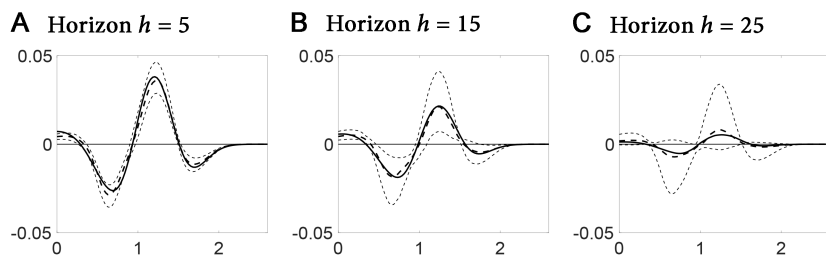


FIG. 6.—Density-differential response. Impulse responses are generated from the estimated VAR. Results are based on $T = 150$, $N = 10,000$, $K = 8$, and $\hat{\lambda}(K)$; see table 3. Solid lines represent “true” responses from the KS economy and dashed bands responses from the estimated functional state-space model (pointwise median and 5th and 95th percentiles).

high-wealth households has to drop. Second, the bands from the percentile VAR are much wider than the bands from the functional VAR.

We provide some intuition for the imprecision of the percentile VAR, using a local projection argument and making a number of simplifying assumptions. A formal analysis in the state-space/VAR setting is beyond the scope of this paper. The functional model implies that the information from the repeated cross sections $X_{i:N,t}$ can be summarized in the K -dimensional vectors $\hat{\alpha}_t$, which are noisy measures of the α_t vectors. The latter evolve jointly with the aggregate variables Y_t according to a VAR. Suppose that the lag order $p = 1$ and that the object of interest is the contemporaneous response to the technology innovation $\epsilon_{z,t}$. Then, write

$$\alpha_t = \Psi \epsilon_{z,t} + \Phi_{\alpha y} Y_{t-1} + \Phi_{\alpha \alpha} \alpha_{t-1} + \tilde{u}_{\alpha,t}, \quad (29)$$

where $\tilde{u}_{\alpha,t}$ is a vector of forecaster errors generated by shocks other than $\epsilon_{z,t}$. We proceed with the following simplifying assumptions: (i) $\hat{\alpha}_t = \alpha_t$ (no uncertainty in regard to cross-sectional densities), (ii) the technology-shock innovations $\epsilon_{z,t}$ are observed (they can be recovered from the observed

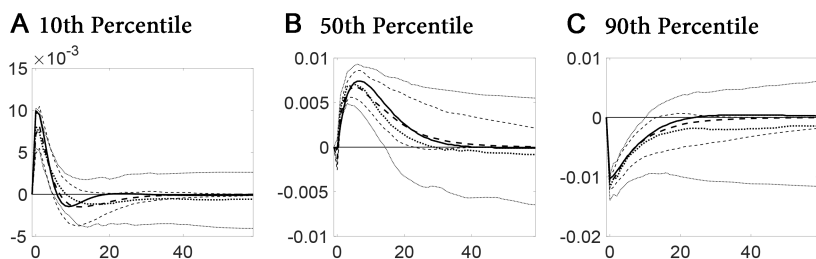


FIG. 7.—Percentile response from functional VAR and percentile VAR: 10th, 50th, and 90th percentiles of the posterior distribution of IRFs. Dashed bands correspond to the functional VAR IRFs and dotted bands to IRFs from the alternative VAR that includes percentiles of the asset/ K distribution. Solid lines represent “true” responses from the KS economy.

TFP process in the KS model and in our application), (iii) the quantiles $q_t = M'\alpha_t$ are a linear function of the α_t 's, (iv) the dimension of q_t is less than K . A linear projection approach based on the functional model would estimate Ψ in equation (29) by OLS, using (Y_{t-1}, α_{t-1}) as controls. Giving this estimator a quasi-Bayesian interpretation, the posterior variance for the impact response is given by $M'(\Sigma_{\tilde{u}_\alpha} \otimes (\sigma_{\epsilon}^2)^{-1})M$, because $\epsilon_{z,t}$ is orthogonal to (Y_{t-1}, α_{t-1}) , where $\Sigma_{\tilde{u}_\alpha}$ is the covariance matrix of $\tilde{u}_{\alpha,t}$ and σ_{ϵ}^2 is the variance of $\epsilon_{z,t}$.

Define $M_\perp$ as a matrix with columns that are orthogonal to the columns of M , and let $W = ([MM_\perp]')^{-1}$ such that we can write the linear projection regression associated with the percentile model as

$$\begin{aligned} q_t &= M'\Psi\epsilon_{z,t} + M'\Phi_{\alpha y}Y_{t-1} + M'\Phi_{\alpha\alpha}Wq_{t-1} + M'\Phi_{\alpha\alpha}WM'_\perp\alpha_{t-1} + M'\tilde{u}_{\alpha,t} \\ &= M'\Psi\epsilon_{z,t} + M'\Phi_{\alpha y}Y_{t-1} + M'\Phi_{\alpha\alpha}Wq_{t-1} + \nu_{\alpha,t}. \end{aligned} \quad (30)$$

The key difference between equations (30) and (29) is that the omission of $M'_\perp\alpha_{t-1}$ as an additional control variable inflates the regression error variance, generates serial correlation in the error term $\nu_{\alpha,t}$ and thereby raises the estimation uncertainty associated with the impact response $M'\Psi$. In practice, the magnitude of the inefficiency depends on the extent to which the percentiles q_t span the information in α_t . But the larger the dimension of q_t , the more likely it is that quantiles may cross in response to shocks, which is clearly an undesirable feature of a “many-quantiles” specification. The inefficiency and the incoherence of a “many-quantiles” model are avoided by our functional approach.

VI. Empirical Analysis

The empirical analysis focuses on the joint dynamics of TFP, real per capita GDP, and employment at the aggregate level and the cross-sectional distribution of earnings. Estimation results for the functional state-space model are presented in section VI.A. In section VI.B, we report impulse responses of aggregate variables, the cross-sectional distribution, and inequality measures derived from the cross-sectional distribution to aggregate shocks. Section VI.C compares IRFs from the functional model to IRFs from VARs that include inequality statistics directly. Finally, in section VI.D, we report responses to shocks that explain the largest share of the variance of particular variables or statistics.

A. Data and Model Estimation

1. Data

We use three macroeconomic aggregates in our empirical analysis: TFP, real per capita GDP, and the employment rate. In addition, we use cross-sectional

data on earnings. Real per capita GDP (A939RX0Q048SBEA) is provided by the Federal Reserve Bank of St. Louis' FRED database, and the TFP series (dtfp) is obtained from Fernald (2012). Weekly earnings (PRERNWA) are obtained from the monthly Current Population Survey (CPS) through the website of the National Bureau of Economic Research (NBER). Weekly earnings are scaled to annual earnings by multiplying by 52. On the basis of the CPS variable PREXPLF ("Experienced Labor Force Employment"), we construct an employment indicator that is 1 if the individual is employed and zero otherwise. This indicator is used to compute the aggregate employment rate.

In the remainder of this paper, we standardize individual-level earnings by $2/3$ of nominal per capita GDP to remove (most of) the common trend from the cross-sectional data. The factor $2/3$ is a rule-of-thumb number for the labor share. After this standardization, an individual earning "1" receives the labor share of GDP per capita. Rather than taking a logarithmic transformation of the standardized earnings data, we apply an inverse hyperbolic sign transformation $x = g(z)$, where x corresponds to the transformed data for which we fit the log-spline densities and z to the original earnings/GDP data. For small values of z , the function is approximately equal to z , and for large values of z , it is equal to $\ln(z) + \ln(2)$. Further details are provided in the appendix.

2. Density Estimation

We take the time period t to be a quarter. For each t , we estimate a cross-sectional density for the transformed earnings-to-GDP ratio. As in the simulation study in section V, we consider different approximation orders K . We use the spline basis in equation (21) and place the knots at predetermined percentiles of the empirical distribution of the x_{it} 's pooled across i and t ; see table 2.

In figure 8A, we show three types of density estimates for the transformed earnings in 2010:Q1: log-spline estimates for $K = 10$ and $K = 22$ and histograms. While the $K = 10$ density estimates are smooth, the $K = 22$ estimates capture the jaggedness of the histograms. The log-spline density estimates are constructed to extrapolate the top-coded income values (spike in the right tail of the histogram). In figures 8B and 8C, we overlay the sample percentiles of the earnings/GDP distribution and percentiles computed from the log-spline density estimates $p^{(K)}(x|\hat{\alpha}_t)$ for $K = 10$ and $K = 22$. The earnings/GDP distribution has a point mass at zero, representing the unemployed individuals, and a continuous part, representing individuals with nonzero earnings. We normalize the estimated density $p^{(K)}(x|\hat{\alpha}_t)$ so that it integrates to the fraction of employed individuals in the sample and apply the change-of-variable formula to convert the density for transformed earnings $x = g(z)$ into actual earnings z .

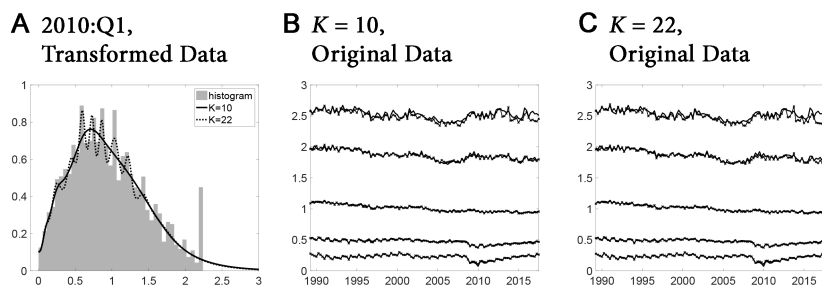


FIG. 8.—Fitted densities and percentiles of earnings/GDP distribution. *A*, Fitted cross-sectional densities and histograms. *B*, *C*, Sample percentiles (10th, 20th, 50th, 80th, and 90th) are shown by plain lines and percentiles from estimated densities by lines with crosses.

Overall, the estimated cross-sectional densities are able to reproduce the time path of the empirical percentiles well, except for some small inaccuracies in the 90th percentile toward the end of the sample. The estimated percentiles for the two specifications are virtually indistinguishable, despite the difference in the density plot in figure 8*A*. Median earnings (relative to the approximate labor share of per capita GDP) fall steadily from above 1 in 1990 to below 1 in 2017. A similar pattern is observed for the 80th percentile. During the Great Recession, there is a noticeable drop of earnings at the 10th and 20th percentiles, followed by a slow and steady rise from 2010 to 2017. Earnings at the 80th and 90th percentiles rise during the Great Recession relative to their 2005 levels.

3. Functional State-Space-Model Estimation

The log-spline density estimation yields a sequence of coefficient vectors $\{\hat{\alpha}_t^{(K)}\}$. The vector Y_t of aggregate variables is composed of TFP growth, real per capita GDP growth, and the (un)employment rate from the CPS data. We set Y_* and α_* equal to the means of these series. After computation of growth rates for TFP and GDP, our sample ranges from 1989:Q2 to 2017:Q3. We consider 1990:Q2 as period $t = 1$ and use the earlier observations to initialize lags in the VAR law of motion.

4. Hyperparameter Selection and Granger Causality

We proceed by evaluating the (approximate) log MDD as a function of the model dimension $K \in \{4, 6, 8, 10, 14, 18, 22\}$, the lag length $p \in \{1, 2, 3, 4\}$, and the hyperparameters λ_1 and λ_2 . For each $\ln \lambda_j$, we consider 31 equally spaced values on the interval $[-10, 20]$. Results are summarized in table 4. For each value of K , we report the optimal $\hat{\lambda}_1$ and $\hat{\lambda}_2$, the number of lags ($\hat{p}$), and the log MDD differentials for $K(\hat{\lambda})$ relative to $K = 4$. With respect to K , the log MDD is maximized for the largest value considered,

TABLE 4
HYPERPARAMETER ESTIMATES AND LOG MDD DIFFERENTIALS

<i>K</i>	OPTIMAL				$\lambda_{2=1}$		
	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{p}$	MDD	$\hat{\lambda}_1$	$\hat{p}$	MDD
4	54.6	1	4	0	54.6	4	0
6	54.6	2.7	4	8,483	54.6	4	8,481
8	148.4	1	4	9,302	148.4	4	9,302
10	54.6	1,096.6	1	9,347	148.4	1	9,330
14	148.4	4.9E-8	1	9,668	148.4	1	9,650
18	403.4	20.1	1	9,711	403.4	1	9,704
22	403.4	7.4	1	12,740	403.4	1	12,730

NOTE.—The log MDD differentials are computed with respect to $K = 4$, $\lambda_1 = \hat{\lambda}_1$, $\lambda_2 = \hat{\lambda}_2$, and $p = \hat{p}$. For each K , we maximized the MDD with respect to λ and p to obtain $\hat{\lambda}_j(K)$ and $\hat{p}(K)$. See the last paragraph of sec. VI.A.4 for the emphasis on $K = 10$ and 22.

$K = 22$. That is consistent with the visual impression from figure 8. The additional knots are used to capture the jagged pattern of the histograms, and the improvement in fit still outweighs the dimensionality penalty induced by the MDD.

The overall degree of selected shrinkage captured by $\hat{\lambda}_1(K)$ is weakly increasing in the dimensionality K . The only exception is the transition from $K = 8$ to $K = 10$, when the lags drop from four to one. Recall that λ_2 controls the relative precision of the prior for the matrix partitions $\Phi_{j\alpha}^{(j)}$ that capture spillovers from the lagged distributions to the current aggregate variables. The optimal values $\hat{\lambda}_2(K)$ show additional shrinkage of the $\Phi_{j\alpha}$ blocks to zero. The degree of relative shrinkage is quite sensitive to K and, except for the $K = 14$ case, not as strong as that in the KS simulation in section V.B. The last column of the table reports log MDD differentials obtained by setting $\lambda_2 = 1$. In this case, the degree of shrinkage for the off-diagonal block $\Phi_{j\alpha}$ is same as for the other coefficients. The MDDs deteriorate, except for $K = 4$ and $K = 8$ when $\lambda_2 = 1$, but the change in MDD is smaller than for changes in K , because λ_2 affects primarily the fit in the time-series dimension, instead of the cross-sectional dimension. Overall, we conclude that the Granger-causal relationship from the cross-sectional income distribution to the aggregate variables is weak, albeit not fully absent.²³

The mechanical application of the MDD criterion suggests proceeding with $K = 22$. However, we are concerned that the jagged pattern of the histograms that the $K = 22$ (and higher) specification is approximating—see figure 8—is more an artifact of the data collection (e.g., survey respondents rounding their earnings) than a genuine feature of the earnings distribution. Moreover, because the earnings are standardized by the

²³ If we restrict the lag lengths to $p = 1$ for $K = 4, 6$, and 8, the relative shrinkage on the off-diagonal blocks increases to 148.4, 54.6, and 20.1, respectively.

continuously evolving GDP per capita, in terms of x -coordinates, the spikes shift from period to period, inducing spurious dynamics. Thus, in the remainder of this section, we first present results for $K = 10$, which delivers smooth estimates of the cross-sectional densities. Results for $K = 22$ are presented in the appendix. While the impulse responses of the cross-sectional densities computed for $K = 22$ inherit the sawtooth pattern visible in the density estimates, the responses for percentiles and inequality statistics derived from the density responses are indeed very similar for $K = 10$ and $K = 22$.

B. Effects of Aggregate Shocks

1. Identification

In the vector Y_t , we order TFP growth first, GDP growth second, and the employment rate third. Let Σ_u be the lower-triangular Cholesky factor of Σ such that $\Sigma = \Sigma_u \Sigma_u'$, and let Ω be an orthonormal matrix. The relationship between the reduced-form innovations u_t and the structural innovations ϵ_t is given by

$$u_t = \Sigma_u \Omega \epsilon_t. \quad (31)$$

It is well known that Ω is not identifiable from the data. Denote the j th column of Ω by Ω_j . We label the first structural shock as technology shock and assume that it is the only shock that affects TFP contemporaneously. Thus, $\Omega_{\cdot 1} = \iota_1$, where ι_j is an $n_y \times 1$ vector in which the j th element is 1 and all other elements are 0. Moreover, we let $\Omega_j = \iota_j$ for $j = 2, 3$. We refer to shocks $\epsilon_{1,t}$, $\epsilon_{2,t}$, and $\epsilon_{3,t}$ as aggregate shocks because they do not affect the cross-sectional distribution contemporaneously. The shocks $\epsilon_{2,t}$ and $\epsilon_{3,t}$ do not have a strict economic interpretation. We refer to them as shocks to GDP growth and the employment rate, respectively. Much of the subsequent discussion will focus on the propagation of technology shocks.

2. Response of Aggregate Variables

Percentiles (10th, 50th, 90th) of the posterior distribution of IRFs are plotted in figure 9. Because the distributional responses are generally small, we scale the IRFs by a factor of 3 and consider 3-SD shocks throughout this section. We compare responses based on the functional model against responses obtained by estimating a VAR in the three aggregate variables only, henceforth “aggregate VAR.” We set $p = 1$ to match the lag length of the functional model and choose the hyperparameter λ_1 by MDD maximization. First, the TFP growth shock raises the level of

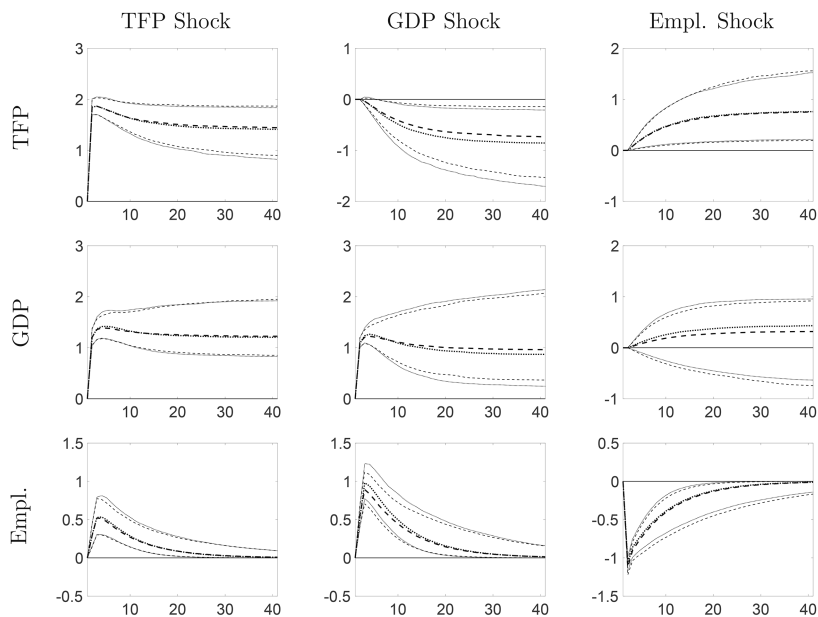


FIG. 9.—Responses of aggregate variables to aggregate shocks: 10th, 50th, and 90th percentiles of the posterior distribution of IRFs for 3-SD aggregate shocks (orthogonalized via Cholesky factorization; see eq. [31]). Panels depict responses of the log level of TFP and GDP, scaled by 100, and responses of the employment rate (Empl.) in percentages. Dashed bands are based on functional VAR(1) with $K = 10$, $\hat{\lambda}_1 = 54.6$, and $\hat{\lambda}_2 = 1,096.6$; dotted bands are based on aggregate VAR(1) with $\hat{\lambda}_1 = 20.1$.

TFP permanently. GDP also rises permanently, and employment shows a positive response (real business-cycle instead of New Keynesian dynamics). Second, the GDP growth shock raises GDP permanently and creates a temporary employment boom and a drop in measured TFP in the long run. Finally, the third shock leads to a drop in the employment rate, and it raises TFP and GDP with a 1-period delay.

The estimated responses from the functional and aggregate models are essentially identical. The degree of IRF overlap is as strong as that in figure 5A, which was obtained from the baseline specification of the KS model that features perfect aggregation. This is consistent with the strong shrinkage of $\Phi_{y\alpha}$ to zero ($\hat{\lambda}_2 = 1,096.6$) reported in table 4. Our empirical result is consistent with Bayer, Born, and Luetticke (2024), who find that including or excluding the inequality data does not change the estimates and variance decompositions of their HANK model. In addition, the aggregation results in Werning (2015) and Berger, Bocola, and Dovis (2023) are compatible with the strong overlap we find.

3. Distributional Responses

Figure 10 illustrates the response of the density of transformed earnings to a 3-SD TFP innovation. As in figure 6, we show the differentials between the shocked density and the steady-state density. Because the employment rate rises in response to a technology shock, the area under the density-differential function is positive. According to the median response, the mass of individuals earning less than the labor share of GDP per capita increases substantially, and initially there is a slight drop in the mass of individuals earning between 1.3 and 2 (in the x -axis of the figure). On impact, the 80% bands are wide and include both positive and negative responses. For horizons $h = 4$ and 12 and earnings below GDP per capita, the density-differential bands include mostly positive values.

Most of the probability mass is added between 0.5 and 1.0. This mass comes from two directions: first, unemployed individuals who find jobs and receive strictly positive instead of zero labor earnings and, second, individuals whose earnings do not rise as strongly as GDP per capita. The first effect is consistent with a model in which individuals are heterogeneous with respect to their idiosyncratic productivity and only individuals whose productivity exceeds a state-dependent threshold work; see, for instance, Chang and Kim (2006) and Chang et al. (2019). In response to an expansionary TFP shock, previously unemployed low-productivity individuals are hired. If wages per efficiency unit are constant, these individuals are likely to earn less than GDP per capita. The second effect requires adjustments on the intensive margin or some heterogeneity in wages per efficiency unit. A strong wealth effect that leads wealthy individuals to reduce their hours or a relative fall of efficiency-unit-specific wages for some individuals is required to shift their earnings from above 1.0 to below 1.0.

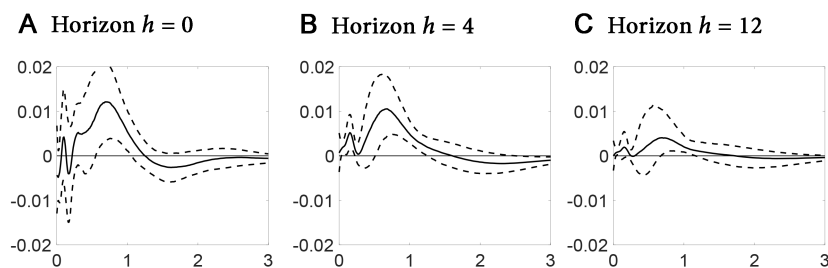


FIG. 10.—Earnings density-differential (transformed data) response to a TFP shock: responses to a 3-SD shock to TFP for $K = 10$. The system is in steady state at $h = -1$, and the shock occurs at $h = 0$. The plots depict the 10th and 90th percentiles (dashed bands) and the 50th percentile (solid lines) of the posterior distributions. We depict differences between the shocked and the steady-state cross-sectional density at various horizons.

4. Inequality Measures

A key advantage of the functional modeling approach is that in any period t the cross-sectional earnings density fully summarizes the earnings distribution. On the basis of the impulse response of the cross-sectional density, we can now compute IRFs of various summary statistics. Figure 11 shows responses for the fraction of individuals with an earnings-to-per-capita-GDP ratio less than 1, the Gini coefficient, and the 10th, 50th, and 90th percentiles of the distribution. The impulse responses are computed relative to an average level of these statistics, indicated by a solid line. The top-left panel reproduces the employment responses. The statistics in the top two rows are computed after assignment of zero earnings to the unemployed individuals. The statistics in the bottom row are computed from the continuous part of the cross-sectional distribution only.

We first consider the “point mass at 0” case. According to the posterior median of the IRF, overall the fraction of individuals earning less than per capita GDP increases from 43.5% to 44%. While the density-differential

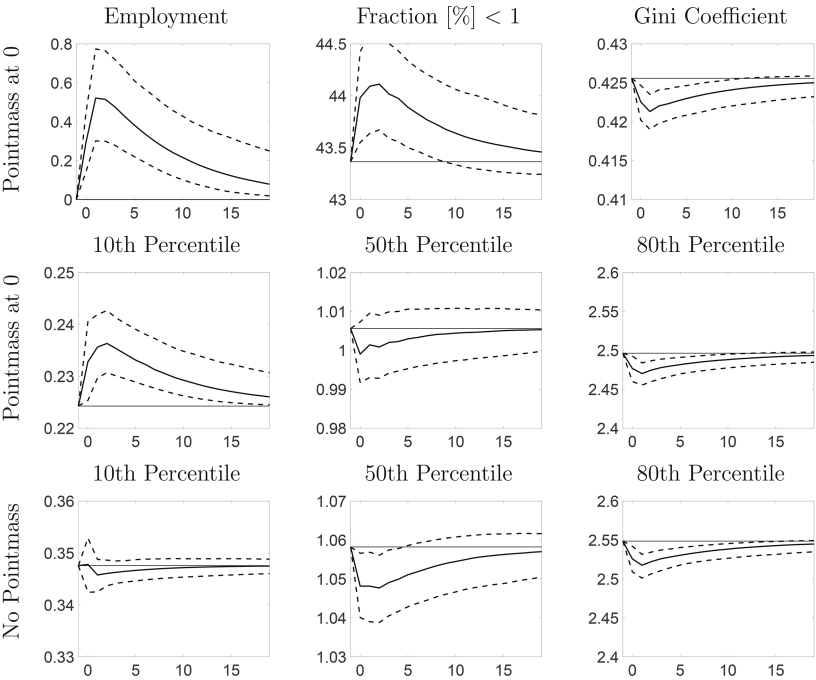


FIG. 11.—Inequality measure (original data) responses to a TFP shock: responses to a 3-SD shock to TFP for $K = 10$. The system is in steady state at $h = -1$, and the shock occurs at $h = 0$. The plots depict the 10th and 90th percentiles (dashed bands) and the 50th percentile (solid lines) of the posterior distributions.

response for $h = 0$ depicted in figure 10 looked “insignificant,” when converted into the fraction earning less than per capita GDP, the response is positive with a probability of almost 90% even on impact. As individuals move from being unemployed (zero earnings) to being employed (positive earnings), the Gini coefficient falls from 0.4256 to 0.4225. Earnings at the 10th percentile of the cross-sectional distribution also increase from 0.224 to 0.233; at the median, they fall slightly, from 1.005 to 0.999, with a band that ranges upon impact from approximately 0.99 to 1.01. Earnings at the 90th percentile fall very slightly, from 2.496 to 2.477. This compression of the earnings distribution is consistent with the decrease in the Gini coefficient.

A comparison between the percentile responses in the second and third rows shows the roles played by the extensive margin, that is, individuals moving into and out of unemployment. Ignoring the point mass at zero raises the baseline level of the percentiles. If the point mass at zero is included, the 10th percentile rises (posterior median) from its baseline level 0.225 to 0.237. If we condition on employment, then the posterior median of the 10th percentile response falls slightly below the baseline level. This is consistent with individuals transitioning into employment after a positive TFP shock generally being low-productivity individuals. The responses of the 50th and 90th percentiles are not sensitive to the extensive margin. As a consequence, not accounting for a point mass at zero dampens the responses of the Gini coefficient and the fraction of units earning less than GDP per capita (not shown in the figure).

C. Comparison to VARs with Inequality Statistics

The advantage of the functional approach is that once the dynamics of the cross-sectional densities have been estimated, it is straightforward to compute the dynamics of any statistic associated with the densities. As a robustness check, we estimate two VARs that combine the three aggregate variables with inequality statistics. We refer to these VARs as “inequality VARs.” Specification 1 combines the aggregate variables with the fraction of individuals earning less than GDP per capita and the Gini coefficient series directly computed from the cross-sectional observations. Specification 2 augments the aggregate variables with the 10th, 20th, 50th, 80th, and 90th percentiles. The inequality VARs also take the form of equation (22), where W_i is redefined to replace α_i with the relevant distributional statistics. The VARs are estimated with the same prior as the functional model. The MDD criterion selects a lag length of $\hat{p} = 4$ and a relative-shrinkage hyperparameter $\hat{\lambda}_2 = 4.9\text{E}-8$, meaning that the estimated system is block triangular and that the inequality statistics do not Granger-cause the aggregate variables. In fact, the $\hat{\lambda}$ pattern is similar to the one obtained from the functional VAR applied to data simulated from the KS economy. In a

similar vein, Berger, Bocola, and DAVIS (2023) construct preference shocks that summarize all the information from the cross section relevant for aggregate dynamics. These shocks' feedback to aggregate variables is found to be quantitatively small.

Figure 12 overlays impulse responses to a 3-SD technology shock for TFP from the functional model and the inequality VAR. The top row contains results for specification 1 and the remaining two rows results for specification 2. The TFP responses between the functional model and the inequality VARs are well aligned, albeit at the posterior median, the long-run response is slightly lower under the former than under the latter.

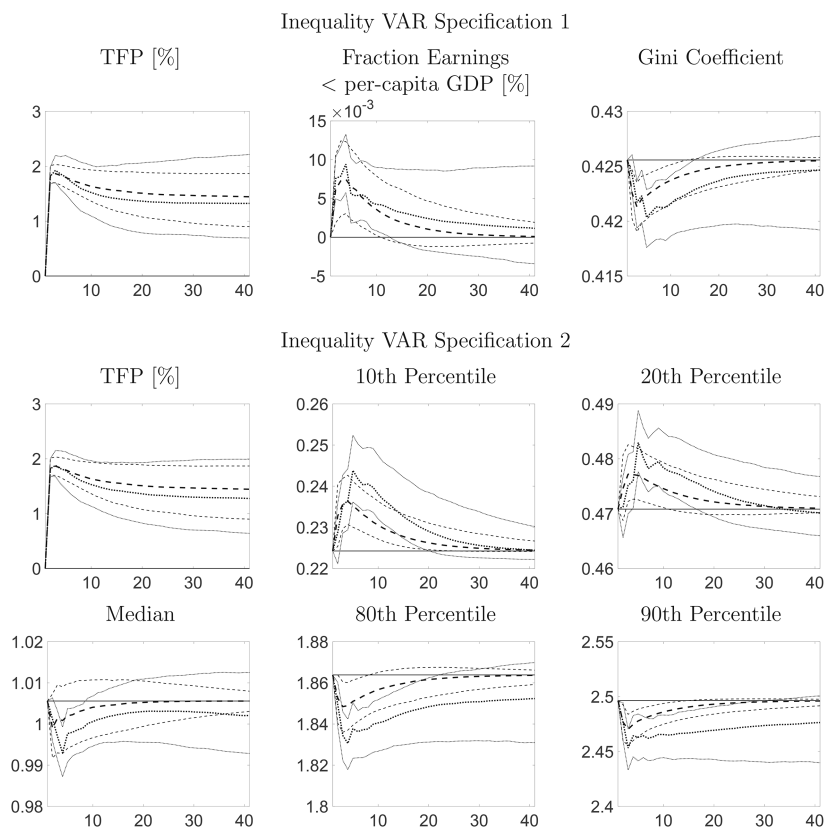


FIG. 12.—Functional model versus inequality VAR: responses to a 3-SD shock to TFP. The system is in steady state at $h = -1$, and the shock occurs at $h = 0$. The plot depicts the 10th, 50th, and 90th percentiles of the posterior distribution. Dashed bands are based on the functional state-space model for $K = 10$; dotted bands are based on the alternative VAR(4). Percentile responses are computed for the original data.

All other responses match in shape, but at the posterior mean the inequality VAR responses are of larger magnitude than the responses computed from the functional model. Most notably, the credible bands for the inequality VARs are much wider than those for the functional model, in particular in the long run. Thus, the functional approach allows for a much sharper inference, despite a higher but still parsimonious parameter count (10 series instead of two or five series to represent the dynamics of the cross section), and is therefore preferable. The empirical findings in regard to the width of the credible bands are consistent with the results from the simulation experiment reported in figure 7.

D. Maximum-Share-of-Variance Shocks

We now consider four shocks that explain the largest share of the variance of a particular variable or statistic. This approach dates back to Faust (1998) and Uhlig (2003) and has been recently used, for instance, to study the anatomy of business cycles in Angeletos, Collard, and Dellas (2020). Recall the relationship between one-step-ahead forecast errors u_t and structural shocks ϵ_t in equation (31). Denote the first column of Ω by q . Then, $\Sigma_w q$ is the impact of a 1-SD shock $\epsilon_{1,t}$. For the first two shocks considered, we choose the value q_* such that the shock $\epsilon_{1,t}$ explains the largest share of the variance of a targeted element of W_t , denoted by $M'W_t$, where M' is a row vector that picks either output growth or employment. Let $\Sigma_w(q; \Phi, \Sigma)$ denote the unconditional covariance matrix of W_t explained by $\epsilon_{1,t}$. Then,

$$q_*(M) = \operatorname{argmax}_{\|q\|=1} M' \Sigma_w(q; \Phi, \Sigma) M. \quad (32)$$

The third shock, henceforth “distribution shock,” targets the variation in the cross-sectional distribution. The unnormalized log density of the cross-sectional observations is given by $\ell_t^{(K)}(x) = \zeta'(x)\alpha_t$. We denote the covariance kernel of this function by $\Omega_\ell(x, \tilde{x}) = \zeta'(x)\Sigma_\alpha(q)\zeta(\tilde{x})$, where $\Sigma_\alpha(q)$ is defined in the same way as $M'\Sigma_w(q)M$ above. Now M' is a matrix that selects the α_t elements from W_t . We discretize the domain of x and then maximize the sum of the eigenvalues of the covariance matrix for the discretized x .²⁴ The fourth shock, henceforth “Gini shock,” is constructed as distributional shock that, conditional on (Φ, Σ) , maximizes the variation in the Gini coefficient without affecting the aggregate variables upon impact. We restrict the candidate values of q to $q = [0', q'_\alpha]'$, where the partition of q conforms with the partition of $u'_t = [u'_{y,t}, u'_{\alpha,t}]$ and q_α is a K -dimensional unit-length vector.

²⁴ Maximizing the trace of $\Omega_\ell(x, \tilde{x})$ produced similar IRFs.

We plot the IRFs for the four maximum-share-of-variance shocks in figure 13. The responses to the shocks that target GDP and employment variance look qualitatively and quantitatively very similar. They mimic the responses to a TFP shock in figure 9. Thus, at the aggregate level, most of the fluctuations in GDP and employment are driven by a TFP-like shock that generates positive comovements between GDP and employment. This shock spills over into the cross-sectional distribution of earnings as more individuals become employed and transition from zero to positive earnings. Earnings relative to GDP rise at the 10th percentile and fall at the 90th percentile, and overall inequality as measured by the Gini coefficient falls.

The IRFs for the shocks that target the share of variance of features of the cross-sectional distribution have effects on the cross-sectional distribution similar to those for the aggregate target shocks: the Gini coefficient

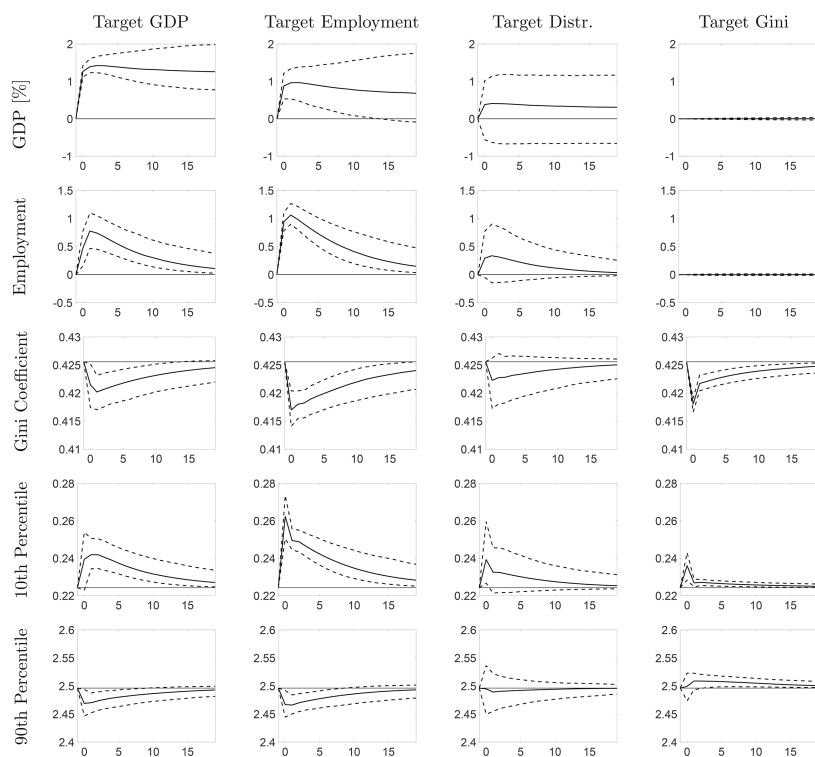


FIG. 13.—Business-cycle anatomy: 10th and 90th percentiles (dashed bands) and 50th percentile (solid lines) of the posterior distributions of IRFs. Sign normalizations: positive response upon impact of target variable for “GDP” and “employment” shock; positive response upon impact of 10th percentile to “distribution” shock (Distr.); and negative response of Gini coefficient to “Gini” shock. The range of the y-axis is identical within rows.

falls, and income at the 10th percentile rises. In response to the distribution shock employment rises, but not as strongly as in response to the GDP and employment-target shocks. The response of output at the posterior median is positive, but the credible bands are wide, and the sign of the response is ambiguous. Finally, the fourth shock is a pure distributional shock that has no effect on aggregate GDP and employment.

A distributional shock could reflect, for instance, an unanticipated revenue-neutral change in fiscal policy that triggers a redistribution of earnings or a change in the underlying idiosyncratic earnings processes, such as an increase in earnings risk. The aggregate effect depends on the nature of the distributional shock. To the extent that the marginal propensity to consume is negatively correlated with income, higher inequality may lower aggregate consumption. If inequality comes from a rise in idiosyncratic volatility, then it could raise precautionary savings. Auclert and Rognlie (2020) show that in general equilibrium a falling interest rate may weaken the negative relationship between inequality and aggregate output and that the net effect is sensitive to monetary and fiscal policy. The absence of an aggregate response to the Gini shock is consistent with the response to a distributional shock that Auclert and Rognlie (2020) find for standard neoclassical models. The quantitatively small effect of a distributional shock to aggregate variables is also consistent with the results of Berger, Bocola, and Dovis (2023) and Bilbiie, Primiceri, and Tambalotti (2023). A thorough assessment of the extent to which the current generation of HA models can reproduce the impulse-response patterns estimated with our functional model is left for future research.

VII. Conclusion

We developed a functional state-space model that stacks macroeconomic aggregates and cross-sectional distributions to provide semistructural evidence about the interaction of aggregate and distributional dynamics. We documented that the model estimated on simulated data is able to reproduce the aggregate and distributional impulse-response dynamics of the underlying KS model economy. In our empirical analysis, we found that adding the earnings distribution to a VAR in TFP growth, GDP growth, and employment hardly affects the estimated propagation of aggregate variables to aggregate shocks, which is consistent with model-based findings reported in Krusell and Smith (1998). We find that an expansionary TFP shock decreases earnings inequality in our sample because it raises earnings at the bottom of the earnings distribution. Finally, we show that the responses of aggregate output to distributional shocks are not significant. We expect the techniques developed in this paper to be widely applicable to studying the interaction between macroeconomic aggregates and cross-sectional distributions and useful for the evaluation of

the most recent vintage of HA models. Extensions left for future work include the introduction of time-varying volatility and allowing for a more general mixed-frequency structure.

Data and Code Availability

Code replicating the tables and figures in this article can be found in Chang, Chen, and Schorfheide (2024) in the Harvard Dataverse, <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/HNPEJB>.

References

- Acharya, Sushant, William Chen, Marco Del Negro, Keshav Dogra, Ethan Matlin, and Reza Sarfati. 2019. "Estimating HANK: Macro Time Series and Micro Moments." Working paper, Fed. Reserve Bank, New York.
- Ahn, SeHyoung, Greg Kaplan, Benjamin Moll, Thomas Winberry, and Christian Wolf. 2018. "When Inequality Matters for Macro and Macro Matters for Inequality." *NBER Macroeconomics Ann.* 32:1–75.
- Amir-Ahmadi, Poyan, and Thorsten Drautzburg. 2021. "Identification and Inference with Ranking Restrictions." *Quantitative Econ.* 12 (1): 1–39.
- Angeletos, George-Marios, Fabrice Collard, and Harris Dellas. 2020. "Business-Cycle Anatomy." *A.E.R.* 110 (10): 3030–70.
- Auclert, Adrien, and Matthew Rognlie. 2020. "Inequality and Aggregate Demand." Working paper, Stanford Univ.
- Bayer, Christian, Benjamin Born, and Ralph Luetticke. 2024. "Shocks, Frictions, and Inequality in US Business Cycles." *A.E.R.* 114 (5): 1211–47.
- Berger, David, Luigi Bocola, and Alessandro Dovis. 2023. "Imperfect Risk Sharing and the Business Cycle." *Q.J.E.* 138 (3): 1765–815.
- Bhandari, Anmol, David Evans, Mikhail Golosov, and Thomas J. Sargent. 2021. "Inequality, Business Cycles, and Monetary-Fiscal Policy." *Econometrica* 89 (6): 2559–99.
- Bilbiie, Florin O., Giorgio Primiceri, and Andrea Tambalotti. 2023. "Inequality and Business Cycles." Working Paper no. 31729 (October), NBER, Cambridge, MA.
- Bosq, Denis. 2000. *Linear Processes in Function Spaces*. New York: Springer.
- Carter, C. K., and R. Kohn. 1994. "On Gibbs Sampling for State Space Models." *Biometrika* 81 (3): 541–53.
- Chan, Joshua C. C. 2022. "Asymmetric Conjugate Priors for Large Bayesian VARs." *Quantitative Econ.* 13 (3): 1145–69.
- Chang, Minsu, Xiaohong Chen, and Frank Schorfheide. 2024. "Replication Code for: 'Heterogeneity and Aggregate Fluctuations.'" Harvard Dataverse, <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/HNPEJB>.
- Chang, Yongsung, and Sun-Bin Kim. 2006. "From Individual to Aggregate Labor Supply: A Quantitative Analysis Based on a Heterogeneous Agent Macroeconomy." *Internat. Econ. Rev.* 47 (1): 1–27.
- Chang, Yongsung, Sun-Bin Kim, Kyooho Kwon, and Richard Rogerson. 2019. "2018 Klein Lecture: Individual and Aggregate Labor Supply in Heterogeneous

- Agent Economies with Intensive and Extensive Margins." *Internat. Econ. Rev.* 60 (1): 3–24.
- Chang, Yongsung, Sun-Bin Kim, and Frank Schorfheide. 2013. "Labor Market Heterogeneity, Aggregation, and the Policy-(In)variance of DSGE Model Parameters." *J. European Econ. Assoc.* 11 (S1): 193–220.
- Chang, Yoosoon, Chang Sik Kim, and Joon Y. Park. 2016. "Nonstationarity in Time Series of State Densities." *J. Econometrics* 192 (1): 152–67.
- Childers, David. 2018. "Solution of Rational Expectations Models with Function Valued States." Manuscript, Carnegie Mellon Univ.
- Cho, Daeha. 2023. "Unemployment Risk, MPC Heterogeneity, and Business Cycles." *Quantitative Econ.* 14 (2): 717–51.
- Coibion, Olivier, Yuriy Gorodnichenko, Lorenz Kueng, and John Silvia. 2017. "Innocent Bystanders? Monetary Policy and Inequality." *J. Monetary Econ.* 88:70–89.
- Del Negro, Marco, and Frank Schorfheide. 2004. "Priors from General Equilibrium Models for VARs." *Internat. Econ. Rev.* 45 (2): 643–73.
- Diebold, Francis X., and Canlin Li. 2006. "Forecasting the Term Structure of Government Bond Yields." *J. Econometrics* 130 (2): 337–64.
- Faust, Jon. 1998. "The Robustness of Identified VAR Conclusions about Money." *Carnegie-Rochester Conference Series Public Policy* 49:207–44.
- Fernald, John. 2012. "A Quarterly, Utilization-Adjusted Series on Total Factor Productivity." Working Paper no. 2012–19, Fed. Reserve Bank, San Francisco.
- Giannone, Domenico, Michele Lenza, and Giorgio E. Primiceri. 2015. "Prior Selection for Vector Autoregressions." *Rev. Econ. and Statis.* 97 (2): 436–51.
- Gordon, N. J., D. J. Salmond, and A. F. M. Smith. 1993. "Novel Approach to Non-linear/Non-Gaussian Bayesian State Estimation." *IEEE Proc. F* 140 (2): 107–13.
- Herbst, Edward P., and Frank Schorfheide. 2015. *Bayesian Estimation of DSGE Models*. Princeton, NJ: Princeton Univ. Press.
- Horváth, Lajos, and Piotr Kokoszka. 2012. *Inference for Functional Data with Applications*. New York: Springer.
- Hu, Bo, and Joon Y. Park. 2017. "Econometric Analysis of Functional Dynamics in the Presence of Persistence." Working paper, Dept. Econ., Indiana Univ.
- Huggett, Mark. 1997. "The One-Sector Growth Model with Idiosyncratic Shocks: Steady States and Dynamics." *J. Monetary Econ.* 39 (3): 385–403.
- Inoue, Atsushi, and Barbara Rossi. 2021. "A New Approach to Measuring Economic Policy Shocks, with an Application to Conventional and Unconventional Monetary Policy." *Quantitative Econ.* 12 (4): 1085–138.
- Kaplan, Greg, and Giovanni L. Violante. 2018. "Microeconomic Heterogeneity and Macroeconomic Shocks." *J. Econ. Perspectives* 32 (3): 167–94.
- . 2022. "The Marginal Propensity to Consume in Heterogeneous Agent Models." *Ann. Rev. Econ.* 14:747–75.
- Koopmanberg, Charles, and Charles J. Stone. 1991. "A Study of Logsplines Density Estimation." *Computational Statis. and Data Analysis* 12 (3): 327–47.
- . 1992. "Logsplines Density Estimation for Censored Data." *J. Computational and Graphical Statis.* 1 (4): 301–28.
- Krueger, D., K. Mitman, and F. Perri. 2016. "Macroeconomics and Household Heterogeneity." In *Handbook of Macroeconomics*, vol. 2A, edited by John B. Taylor and Harald Uhlig, 843–921. Amsterdam: North-Holland.
- Krusell, Per, and Anthony A. Smith Jr. 1998. "Income and Wealth Heterogeneity in the Macroeconomy." *J.P.E.* 106 (5): 867–96.
- Liu, Laura, and Mikkel Plagborg-Møller. 2023. "Full-Information Estimation of Heterogeneous Agent Models Using Macro and Micro Data." *Quantitative Econ.* 14 (1): 1–35.

- Meeks, Roland, and Francesca Monti. 2023. "Heterogeneous Beliefs and the Phillips Curve." *Monetary Econ.* 139:41–54.
- Mongey, Simon, and Jerome Williams. 2017. "Firm Dispersion and Business Cycles: Estimating Aggregate Shocks Using Panel Data." Manuscript, Univ. Chicago.
- Ottonello, Pablo, and Thomas Winberry. 2020. "Financial Heterogeneity and the Investment Channel of Monetary Policy." *Econometrica* 88 (6): 2473–502.
- Ramsey, James O., and Bernard W. Silverman. 2005. *Functional Data Analysis*, 2nd ed. New York: Springer.
- Reiter, Michael. 2009. "Solving Heterogeneous-Agent Models by Projection and Perturbation." *J. Econ. Dynamics and Control* 33 (3): 649–65.
- Schorfheide, Frank, and Dongho Song. 2015. "Real-Time Forecasting with a Mixed-Frequency VAR." *J. Bus. and Econ. Statis.* 33 (3): 366–80.
- Shao, Jun. 1997. "An Asymptotic Theory for Linear Model Selection." *Statistica Sinica* 7 (2): 221–64.
- Uhlig, Harald. 2003. "What Moves Real GNP." Working paper, Univ. Chicago.
- Varin, Cristiano, Nancy Reid, and David Firth. 2011. "An Overview of Composite Likelihood Methods." *Statistica Sinica* 21 (1): 5–42.
- Villalvazo, Sergio. 2021. "Inequality and Asset Prices during Sudden Stops." Working paper, Univ. Pennsylvania.
- Werning, Iván. 2015. "Incomplete Markets and Aggregate Demand." Working Paper no. 21448 (August), NBER, Cambridge, MA.
- Winberry, Thomas. 2018. "A Toolbox for Solving and Estimating Heterogeneous Agent Macro Models." *Quantitative Econ.* 9 (3): 1123–51.

Copyright of Journal of Political Economy is the property of University of Chicago Press and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.